

Application of phase change materials in solar water heating systems- A comprehensive review

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ABSTRACT

Solar water heaters (SWHs) are widely utilised to meet domestic and industrial hot water demands due to their technical feasibility and economic viability. However, their performance is significantly affected by the intermittent nature of solar radiation, limiting energy availability to daylight hours. To address this limitation and ensure round-the-clock hot water supply, integrating Phase Change Materials (PCMs) offers a promising solution by enabling efficient thermal energy storage. This comprehensive review explores the fundamentals and classifications of PCMs, various integration strategies within SWH systems, and advanced techniques aimed at improving their thermal performance. Special emphasis is given to the use of nano-enhanced and hybrid PCMs, which demonstrate superior thermal properties compared to conventional materials. The review further examines experimental and numerical studies comparing different PCMs and system configurations, highlighting performance improvements through the use of nanoparticles, metallic foams, expanded graphite, boron nitride, and finned or encapsulated structures. It is observed that materials such as paraffin wax, fatty acids, sugar alcohols, and hydrated salts are commonly employed due to their suitable phase transition temperatures and thermal characteristics. The article also evaluates the economic feasibility, technical challenges, and future directions of PCM-integrated SWH systems. Selection of appropriate PCM is found to depend on factors like thermal conductivity, charging/discharging rates, operating temperature, and required storage capacity. Overall, the findings indicate that PCMs especially in nano-enhanced or composite forms hold substantial potential to enhance the efficiency, reliability, and sustainability of solar water heating technologies.

1. Introduction

The current era can be referred to as the machine age or the industrial age. Energy is necessary for operating machines, and it may originate from a variety of sources, including those that are renewable and non-renewable, such as petroleum, hydraulics, nuclear power, solar energy, wind energy etc. The rate of population growth has picked up significantly, which will inevitably result in a rise in the need for energy in the future. Currently, most of the energy demand is met by limited non-renewable energy resources, hence current research should concentrate on developing renewable energy resources to meet the upcoming demand. Petroleum is depleting owing to high usage in automobiles, industries, airlines, etc. Hydraulics, nuclear, and wind energy are expensive or need regular maintenance. Fossil fuel emissions induce

acid rain and ozone layer depletion. Every year, over 160,000 individuals worldwide pass away due to direct or indirect climate change-related causes [1]. Solar energy is clean, unlike fossil fuels and emits no greenhouse emissions when used. Solar energy is abundant and free of cost. The environment is not significantly harmed by solar energy, making it an ecologically beneficial source of energy. Solar energy is used in PV cells, air heaters, water heaters, cookers, chimneys, etc. The unending supply of energy from the sun is quite enormous and approximately 1.8×10^{11} MW of solar energy is intercepted by the earth [2]. Solar thermal applications are split into three categories based on the temperature ranges they are used for: (a) Solar drying, hot water supplies, and cooking are examples of low-temperature applications (below 100 °C) (b) Refrigeration, industrial process heat, and other medium-temperature applications (below 150 °C) and (c) Applications

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that require high temperatures (over 150 °C), such as energy generation [3]. Solar water heating devices are used to heat water by utilizing radiation from the sun. The most common use of solar energy is for household water heating across the world. Following the 1973 oil crisis, there was a rise in curiosity in solar water heating systems integrated with flat plate collector. Solar air heaters operate similarly to solar water heaters (SWH), except of water, air is being used as fluid. It is typically used for heating a space throughout, in chilly climates [4]. Sun is necessary for solar-related devices to work, but unfortunately, solar radiation can only be received during the daytime. As a result, the operation of equipment powered by solar radiation energy is intermittent. Intermittence is a fundamental disadvantage of solar energy [5]. Energy storage technology was incorporated by several researchers to address the problem of intermittent operation of solar systems and performance enhancement [6–8]. Solar energy storage technology has been developed to store solar radiation energy in the form of heat or electricity for use by solar-powered devices on foggy, hazy days [9,10]. To operate solar street lights, off-grid solar power plants, and solar pumps during night or cloudy condition, the solar PV systems with battery-based energy storage are employed.

Solar thermal systems equipped with thermal energy storage (TES) units are also put to use for a variety of domestic and residential applications. The energy storage improves the performance of solar devices. Several distinct strategies for the storage of energy have been developed and currently in widespread use [11]. Sensible heat storage, latent heat (or phase change) storage, and chemical heat storage are the three main categories into which thermal energy storage technologies are typically divided according to their thermophysical characteristics, as shown in Fig. 1 [12,13]. Sensible heat storage, which uses temperature changes in solids or liquids to store energy, and latent heat storage, which uses material phase transitions (such as solid–liquid, liquid–vapor, or solid–solid) to store and release energy at almost constant temperatures, are further subcategories of TES in the context of non-reactive storage media. However, chemical heat storage offers a viable path for high energy density and long-duration storage since it uses reversible endothermic and exothermic reactions in reactive materials. A fundamental understanding of TES technologies is given by this classification framework. Of these, latent heat storage employing phase change materials (PCMs) is becoming more and more acknowledged for its exceptional energy density and capacity to stabilize temperature in solar water heating systems.

Solar energy is a time-dependent renewable source of energy that is inexhaustible and eco-friendly. Among the most often used applications of solar energy is for thermal purposes. On the other hand, thermal energy storage is an efficient way to combat the sporadic and diffuse character of solar energy [9]. Techniques for energy storage can work towards closing the gap between supply and demand for energy. Among the most feasible methods for storing solar energy involves the utilization of specific organic and inorganic substances, which are referred to as phase change materials (PCMs), which enable the latent heat of fusion

to be harnessed [14]. To improve the thermal performance of solar heating systems, PCMs can be used as an effective tool. PCMs can effectively store additional thermal energy during the day through fusion and release it at sunset via solidification at a consistent temperature, which leads to a higher storage efficiency. PCM has a number of benefits. Specifically, its high heat storage capabilities compared to other heat storage materials lead to a decrease in the size of the thermal storage facility [15]. The isothermal phase change feature of PCMs, wherein heat is charged and discharged at a virtually fixed temperature (phase change temperature) matching the phase transition, seems to be another significant benefit. As a consequence, they are perfect for applications involving space heating where thermal comfort can be attained by reducing temperature swings. However, PCMs have a number of demerits such as PCM systems necessitating a long operational time to recuperate the installation cost. Conventional PCMs also have other drawbacks such as low thermal stability, thermal cycling, and low thermal conductivity [16]. There has been a number of attempts to review the implementation of PCMs in different solar energy systems, scholars focused on discussing the concepts of these materials with outlining the influential parameters towards its implications in the solar energy industry. That is why over past ten years, the role of PCM in solar water heating are becoming the green field of research and different categories of researches are using this technology “PCM in solar water heating” as per web of science (WOS) core collection database as depicted in Fig. 2 (a & b). To understand the average number of publication on PCM in solar water heating in the Web of science database, the WOS data analysis has been done and it was found that the number of publication on ‘PCM in solar water heating’ are quite good as illustrated in Fig. 2(c) and their application in different fields are increasing day by day. Further, Several researchers from different countries are working in this field and published various literatures on this topic as depicted in Fig. 2(d). These figures proposed that the future of phase change materials in solar water heating are developing drastically and explaining the potential of this field which allows researchers for further studies on future applications.

In the field of solar thermal systems, the TES material has been used in variety of applications for performance improvement of the system. Jawad et al. [17] has examined the performance of solar air heater in Baghdad city with aluminum chip and paraffin wax with SiC nano-composite material as energy storage material. The thermal conductivity rose by 18.2 % when 3 % weight ratio of nano-SiC was incorporated to paraffin. The highest exit air temperature of 64.4 °C was attained between 1 and 3 p.m. It also keeps working for a minimum of 3 h after sundown. The aluminum chip enhanced the solar heater’s effectiveness and postponed the usage of paraffin/nano-SiC storage heat till noon. The solidification and melting points also dropped by 5 % and 5.2 %, respectively. Habib et al. [18] has investigated the solar air heater performance at 3 % SWCNT in paraffin wax energy storage material. It was found that addition of SWCNT in paraffin wax reduced the melting and solidification points of the nano composite PCM by 4.5 °C and 9 °C, respectively, and enhanced the thermal conductivity by 12 %. Compared to pure paraffin, the SWCNT/paraffin nanocomposite increased the thermal energy stored, by 20.7 % in natural convection and 21.2 % in forced convection. Ibrahim et al. [19] has explored that Iraqi paraffin can be used for applications of solar energy by adding hybrid nano-TiO₂ and MgO (50 % each). The maximum increase in thermal conductivity has been recorded as 24.92 % for hybrid paraffin wax nano composite. Chaichan et al. [20] has examined the use of paraffin wax PCM in solar concentrator heater for solar distillation applications and with the addition of PCM and a concentrating dish, for sun tracking, the system productivity, heating efficiency, and concentrating efficiency have all risen by around 307.54 %, 112.87 %, and 64.07 %, respectively, and the system’s operating duration was raised to around 5 h. Also, by removing the solar tracker and adding PCM, the system’s operating duration was extended to 3 h. Elsheikh et al. [21] has investigated the utilization of nanofluid for the performance improvement of various solar

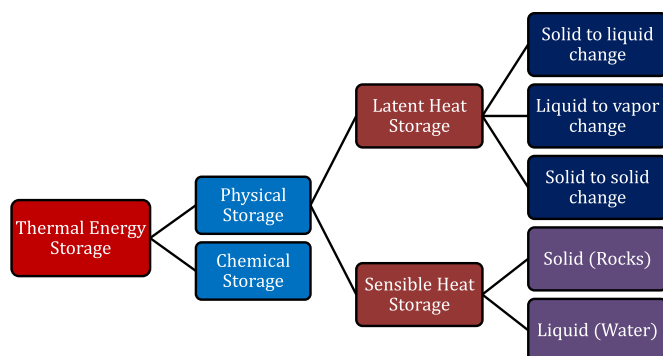


Fig. 1. Categorization of the Solar thermal energy storage techniques.

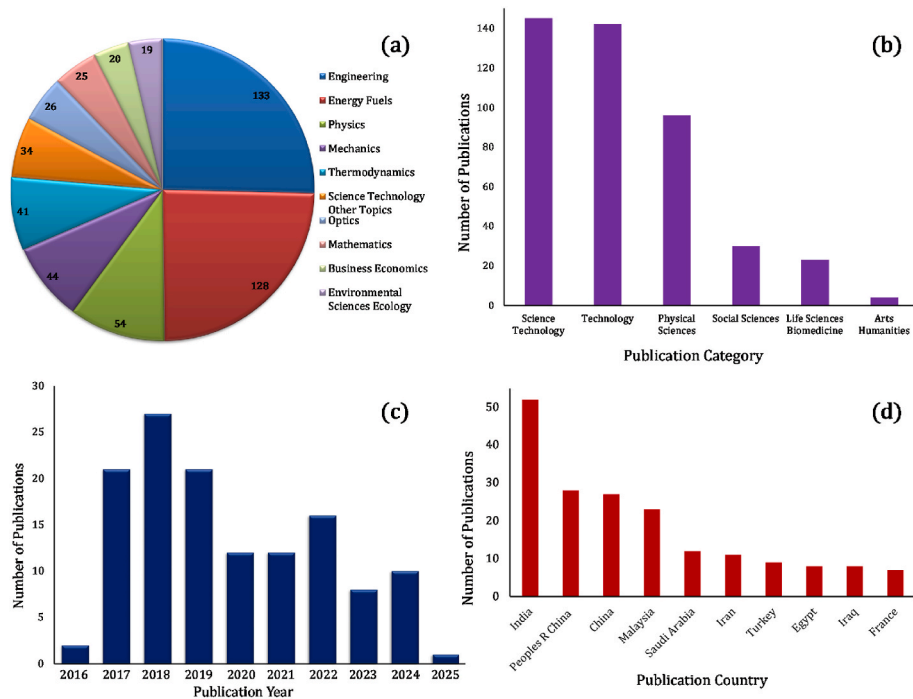


Fig. 2. Publication details on 'PCM in solar water heating' based on WOS database: (a & b) Different fields and categories of research using PCM in SWH, (c) Number of publications on PCM in SWH over past ten years and (d) Top 10 countries for publication on PCM in SWH.

applications like solar water heater, distillation, solar PV/T system etc. Zayed et al. [22] focuses on the PCM container's geometrical arrangements and design factors, with the goal of enhancing TES system performance. The application of metal foams as well as graphite, incorporation of fin, integration of fins with nanoparticle, use of microencapsulated PCMs, and doping of highly photothermal materials are only methods for increasing PCM heat transfer. The primary difficulty in using PCM is the container design and because of its rapid melting rate and storing effectiveness, the rectangular PCM tank is the best container for bulk storage. Vedanarayanan et al. [23] performed research work on evacuated tube solar air heater using stearic acid as PCM and water as base fluid to heat air. This system has an advantage over sensible energy storage methods as it may continue to be utilised after the sunset. This advantage is due to the PCM's superior capacity for storing heat. Jegan et al. [24] discussed the advances PCM and their applications in solar thermal energy storage with better stability, latent heat of fusion and heat capacity. PCM has a greater capacity for absorbing energy than the other materials. Paria et al. [25] studied PCM behavior during solar thermal energy storage charging and discharging and the effect of flow rate on charged and discharged solar thermal energy in finned shell and tube heat exchangers with distilled water as the heat transfer fluid. The findings shown that the overall melting time is decreased by 58 % by raising its Reynolds number from 1000 to 2000. Peng et al. [26] explored the five characteristics of PCM such as PCM classification, microencapsulation technique, encapsulation shell material, encapsulated PCM characterization and different thermal application. The microencapsulation of PCMs used in thermal storage improves their thermal and mechanical efficiency by increasing heat transfer area and reducing the leaking of melting PCM. Mao et al. [27] have done critical analysis of various PCM as per their melting point range and heat storage capacity. A review article has been written on advancements in SWH technology and prospects for future development [28].

In an experiment conducted on the heat storage capacity of evacuated tube solar water heater (ETSWH) using twin nanoparticle 1 % mass (CuO and SiO₂ with equal quantity) in paraffin wax as PCM [29]. Only Paraffin and, twin-nano/paraffin integration increased the heat storage capacity of evacuated tube and temperature of water by 8.7 °C and

11.6 °C respectively, in the next morning. A PCM has been developed using a unique pseudo ionic PCM comprised of urea and cetyltrimethylammonium bromide (CTAB). CTAB-urea with a 6:1 mass ratio had the highest latent heat of 174.9 kJ/kg, 86.9 °C melting temperature, better stability, low supercooling of 3 °C, and good thermal conductivity [30]. In a study a PCM has been prepared with polyethylene glycol based composite with very high heat storage capacity, thermal conductivity and stability, for solar water heating applications [31]. Researchers have investigated a U-type evacuated tube solar collector that uses PCM to store excess solar energy during the daytime and utilize it in night. The optimal charging and discharging tube diameter is 6 mm, and the proposed model improved thermal energy by 20 % [32]. In a work one has added the nano-silica to CaCl₂·6H₂O (an inorganic PCM) to increase its energy storage capacity. The findings showed that adding 0.6 % nano-silica to CaCl₂·6H₂O increased its thermal conductivity without reducing its latent content [33]. In this field, the researchers have studied use of PCM for performance improvement of various solar applications. The low thermal conductivity, tendency to separate into several phases, subcooling/supercooling effects, incompatibility with other materials, and environmental hazards hinder the widespread implementation of PCMs [7]. In a work the investigators have focuses on current developments in PCM integration into various solar collector designs. PCM are excellent and active for long-term heat storage and thermal control [34]. The performance of SWH using PCM exhibits improvement in its overall performance [35]. As per the literatures explored, some researchers have also investigated the effect of PCMs on performance of solar thermal systems. The use of solar energy with advancements in thermal energy storage systems is an emerging field to meet future energy demand. Unfortunately, there is little literature available on the effective use of PCM and current advances in solar water heating systems. To fill the gap, a review has been conducted on basic details of PCMs and their current advanced utilization in solar water heating system. Current work emphasizing on classification of PCM, desirable properties of PCM, suitability of PCM for various temperature range applications, thermal properties of PCM, thermal conductivity enhancement techniques for better charging and discharging cycles, PCM encapsulation, modified rise design, fin based PCM, hybrid nano

PCM, modified evacuated tube collector PCM and use of NCPCM in various applications of solar water heating system. Furthermore, recent applications of PCM with different combinations of nanoparticles and fins for solar water heating application have been comprehensively reviewed.

In a recently conducted investigation of utilization of phase change material in energy storage applications. In this study researcher have done comprehensive review and mainly focuses on utilization of effective used phase change material in solar collector for making it an efficient thermal energy storage [36]. The eutectic phase change materials are also gaining attention as a thermal energy storage material for solar water heat application. In a review article, the thermo physical characteristics, reliability, thermal stability of specifically, eutectic phase change materials have been comprehensively explored to analyse its suitability for the application in solar water heaters [37]. In the recent time phase change material incorporated with nano particles are gaining attention for it's used as an efficient thermal energy storage material for various applications. The nano composite phase change material can also be effectively used in solar water heating applications. The applications of nano enhances phase change material are shown through Fig. 3 [38]. The encapsulation of phase change materials for solar thermal energy storage for enhances system performance is also an emerging field for research. In a study researchers have focused on the encapsulation of PCM for solar thermal applications. The study recommends the encapsulation of PCM can be done many solar thermal systems like, solar water heater, solar desalination, solar pond, etc [39]. Researchers are continuously to find new technique to enhance the performance of PCM based solar thermal energy devices. In this process, recently researchers have explored that metal foam with PCM can enhance the system performance. A comprehensive content is available on this unique technique. The study delineates the models and methodologies now used to characterize the phase transition of phase change materials inside metal foams [40].

In the recent year, many review article has provided the insight of utilization of phase change materials for solar thermal energy storage applications. The review articles have discussed the different aspect of solar thermal energy storage devices for applications like solar chimney, solar still, and solar dryer [41–43]. The past studies show that the phase change materials is also been used for solar waters heaters for thermal energy storage. As per the analysis of available content, it has been obtained that, some researchers have focused on only incorporation of nanomaterial in phase change material in performance enhancement, some have focused only on encapsulation technology of phase change materials, some focused on eutectic phase materials during analysis, few researchers focused on phase change materials with metal foam and some has focused on phase change material in solar collector. As per the survey done, very few literature are available which cover the overall aspect of phase change material utilization in solar water heating system for enhanced and continuous output. Previous studies lacks comprehensive analysis on PCM, its integration techniques, performance improvement, comparative analysis, economic aspect as well as challenges and future outlook of PCM based solar water heater. As per the literature explored regarding PCM, and its application in solar water heaters the primary aims of this investigation can be summarized as follows:

- Investigate the solar thermal energy storage technique, phase change material characteristics classifications, publication details for integration of PCM in solar water heaters.
- Examine the various integration methods of PCM in solar water heaters for enhanced thermal characteristics as well as improved system performance.
- Conduct comparative analysis of PCM and PCM with nano additive with conventional solar water heater and identify key outcome of the study.

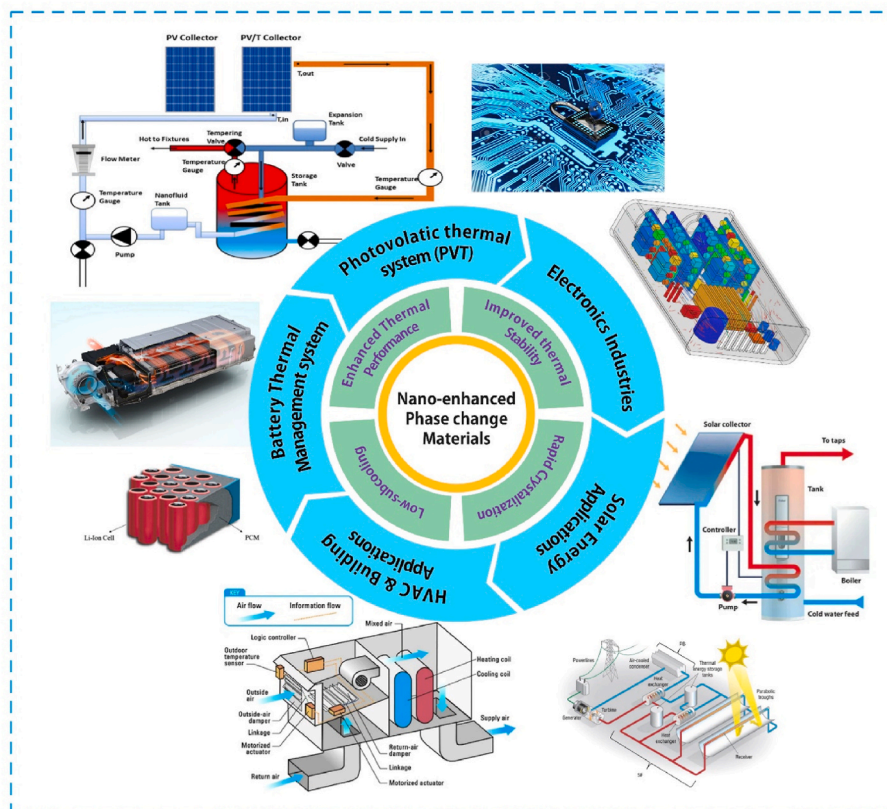


Fig. 3. Applications of PCM incorporated with nanoparticle [38].

- Analyse the costing, challenges associated, future outlook and certain recommendation for PCM based solar water heater

In this comprehensive review article, the study emphasis on the utilization of phase change materials in solar water heaters, particularly regarding phase change material characteristics and its different ways of use in the system. The aspects contributing to the study include:

- The study provides the comprehensive knowledge about PCMs its characteristics, performance analysis of solar water heaters (flat plate collector based, evacuated tube collector based, concentrator based system) with integration of PCM, nano composite PCM, and hybrid nano composite PCM.
- This articles provides the critical analysis of various thermal conductivity augmentation technique for better heat transfer characteristics followed by enhances solar water heater performance.
- This study includes the economic analysis of PCM and nano composite PCM based solar water heater, comparative analysis of cost effectiveness and performance with conventional solar water heaters. The challenges and future outlook of this current study have also been covered.

2. PCM as solar TES

Solar energy is a pristine and environmentally friendly source of energy that has the potential to address the limitations and environmental concerns associated with traditional energy sources. However, the utilization of solar energy necessitates the implementation of energy storage devices to store and then provide the energy for future usage. The implementation of a solar thermal storage system has the potential to enhance the performance of solar thermal applications by effectively managing fluctuations in solar radiation that occur over the course of a day. PCMs are considered to be the suitable material for TES because of their significant latent heat capacity. PCMs has the potential to store a greater amount of heat per unit capacity compared to traditional storage methods, because of their latent heat storage mechanism [44]. One of the efficient method for increasing TES is the utilization of PCMs [18]. A study discussed that PCMs absorb solar radiation and store it as latent heat. The heat storage of PCM occurs appropriately at constant temperature as illustrated in Fig. 4(a) [45,46]. Moreover, heat storage in PCM and its utilization for any purpose is a cyclic process. The cyclic process of the heat energy storage concept of PCM for solar thermal application is shown in Fig. 4(b). As per practical applications, the PCM improves the efficiency of Rankine cycle. The PCMs are also widely used in low temperature TES systems, like building space heating, hot water distribution, and heat sinks for electronic equipment such as laptop computers [47]. PCMs provide an appealing solution to solar thermal storage; nonetheless, it is vital to thoroughly evaluate their long-term

performance, cost-effectiveness, and integration techniques.

2.1. PCM fundamentals and classification

The use of PCMs has garnered significant interest among academics globally, primarily owing to its considerable potential in the field of TES. The broad temperature range capability and compositional adaptability PCMs render them highly suitable for a diverse variety of thermal storage and management systems. To develop a comprehensive PCM with exceptional energy storage capabilities and related properties, it is crucial to conduct a thorough comprehension of PCMs and their classification [49]. The detailed classification of PCMs are illustrated in Fig. 5 (a). Hydrocarbons, such as paraffin, fatty alcohols, and waxes, are used to make organic PCM. An inorganic PCM contains molten salts, salt hydrates, and metallic. Organic-inorganic, inorganic-inorganic, and organic-organic molecules are all included in eutectic PCM. PCMs can be used in numerous ways, such as particles distributed in bulk materials, matrices and sheets [50–52]. As a result of their excellent thermal specifications, organic solid–liquid PCMs are an important group of PCM, with a high TES density, a tiny temperature range of phase change, no or little super-cooling and congruent melting. The use of these PCMs, on the other hand, is limited by leak from the storage tank and of having less thermal conductivity [53]. Organic PCMs have twice of per unit volume latent heat storage capacity, greater heat conductivity and less expensive [47,54]. Inorganic PCMs, on the other hand, have been claimed to exhibit inherent flaws such as incongruent melting, corrosiveness, super cooling and phase segregation characteristics [54]. The fundamental advantage of eutectic mixtures has been discovered to be their phase transition temperature, which can be controlled by mingling the several mass ratios of the elements [55].

In the specified temperature range, solid-solid PCMs maintain their bulk solid characteristics. They don't have the leaking problems that solid-liquid PCMs do, and as a result, non-encapsulated solid-solid PCMs could be employed with different materials [58]. Due to the substantial variations in volume and pressure during change of phase, it has been stated that the use of PCMs in solid–gas and liquid–gas forms is limited [59]. Furthermore, PCMs can be classified as low (less than 220 °C), medium (220–420 °C), and high temperature materials (greater than 420 °C) with melting points [13,60]. Organic PCMs have been shown to be suitable for application having lower temperature requirement [61]. Inorganic salt, (such as chlorides, metal, carbonates fluorides, sulphates, and nitrates) are typically exploited for high-temperature uses [62]. Organic molecules have low melting point (between 10 °C and 60 °C), which makes them suitable for application having lower temperature requirement. The eutectic mixture looks promising at around 100 °C. In the medium temperature range, eutectic mixtures of inorganic compounds heated from 130 °C to 1250 °C are commonly utilised. For high-temperature applications, mixtures of inorganic chemicals such as

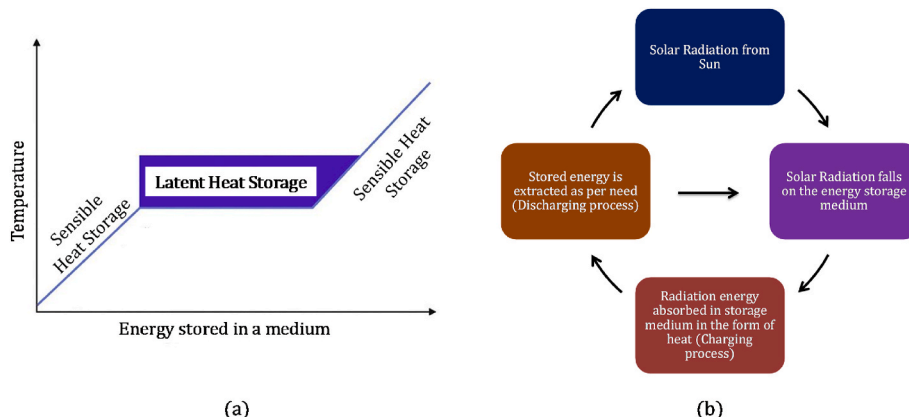


Fig. 4. Graphical representation of (a) Temperature variation with energy storage medium [48] and (b) Charging/discharging cycle of PCM.

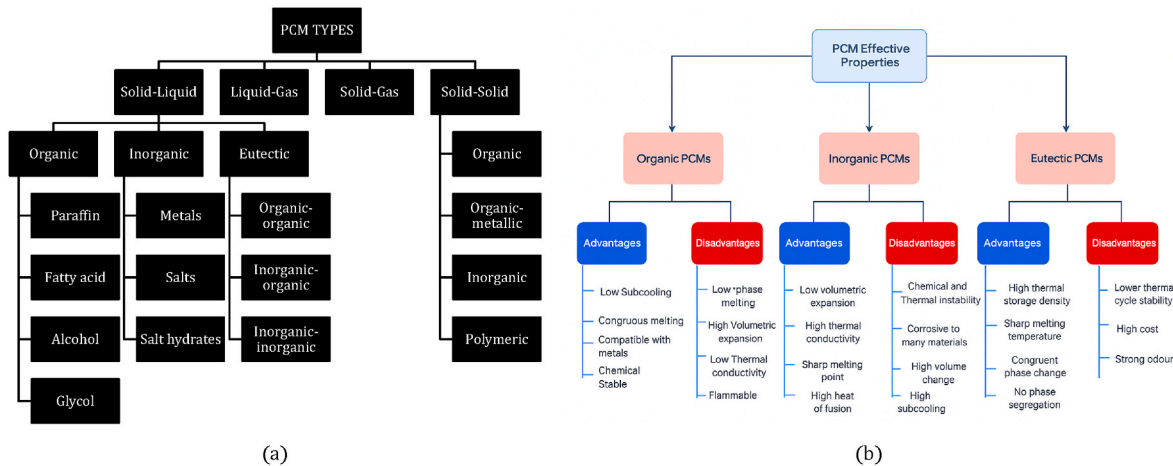


Fig. 5. Pictorial representation of (a) Categorization of different PCMs [56] and (b) Effective properties of PCMs [57].

sodium and potassium that melt about 170 °C have been shown to be beneficial [63]. Metal hydrides have been proven to be an alternative for energy storage in future [64].

2.2. Desirable properties of PCMs for SWH

The various categories of PCMs have their own advantages and disadvantages. Organic PCMs have garnered attention because of their adequate heat storage density, broad temperature operating range, stability over time, noncorrosive, and low toxicity features [65,66]. On the other hand, the organic PCMs that are often used in solar thermal conversions have number of drawbacks, the most frequent of which are naturally poor thermal conductivity, lack of fire safety, and a high price [67,68]. Inorganic salt hydrates have high phase transition enthalpy, are nonflammable, and cost-effective, making them suitable for TES applications. However, these salts often have severe supercooling, phase segregation, and corrosivity problems [69]. The effective properties of various PCMs as per advantages and disadvantages for solar TES are depicted in Fig. 5(b) [57]. Further, PCMs used in solar-powered thermal applications should possess distinct desirable characteristics to maximise TES efficiency. The desirable PCMs properties are shown in Table .1, and thermal properties like melting point and latent heat storage capacity is presented in Table .2.

However, to select an acceptable PCM, the following parameters should be used in the following order:

- a) Latent heat of the material
- b) Material thermal conductivity
- c) Melting point of the material
- d) Specific heat of the material

Table 1
List of desirable properties of PCMs [61].

Desirable properties of PCMs	• Thermo physical properties	• Suitable phase change temperature
		• High value of latent heat of fusion
		• Lower degree of supercooling
		• Low vapor pressure
		• Small change in volume during charging & discharging
		• High thermal conductivity
	• Chemical properties	• Compatibility with other material
		• No chemical decomposition
		• Non-flammable
		• Non-poisonous
		• Non-explosive
	• Economical properties	• Less cost
		• Good recyclability

e) Density of the material

To ensure higher rate of energy storage and discharge, PCM must have a higher value of thermal conductivity, latent heat and a lower value of melting point. In case of imperfect contact between PCM layer and absorber material, the vacant spaces in the PCM layer will produce thermal contact resistance, lowering heat transmission between the layer of PCM and absorber material [78]. The lower cost of storage material lowers both capital and operating costs.

3. Integration strategies and enhancement methods of PCMs in SWH

The integration of PCMs into SWH systems has emerged as a promising approach to enhance thermal energy storage efficiency and extend hot water availability beyond sunlight hours. Recent advancements have focused on strategic PCM placement and encapsulation techniques to optimize heat transfer and thermal regulation. Researchers have explored several configurations, including direct immersion of encapsulated PCMs in the storage tank, incorporation within absorber plates of flat-plate collectors, and integration into thermosyphon systems. Among these, encapsulated PCM systems using spherical, cylindrical, or flat panel geometries demonstrate improved thermal conductivity and phase stability, especially when enhanced with fins, metal foams, or high-conductivity nanoparticles. Shell-and-tube and finned-tube arrangements have also gained attention for their high surface area and efficient heat exchange characteristics. Moreover, cascaded PCM systems, which utilize multiple PCMs with varying melting points, offer superior thermal stratification and broader operational temperature ranges, enabling more consistent water temperatures. Studies have highlighted that the latent heat storage of PCMs significantly reduces temperature fluctuations and enhances the thermal response of SWH systems under intermittent solar conditions [79,80]. For instance, paraffin-based PCMs with melting points in the 50–70 °C range, when encapsulated in aluminum or polymer shells, have shown effective thermal buffering in both laboratory and field tests [81]. Overall, the choice of PCM type, encapsulation strategy, and integration point within the system plays a critical role in determining the long-term reliability, energy savings, and cost-effectiveness of PCM-enhanced solar water heaters.

3.1. Thermal conductivity augmentation technique

The lower value of PCMs thermal conductivity is critical for its implementation because it translates to poor thermal performance in TES systems. Thus, PCMs with better thermal conductivity values are

Table 2
PCMs thermal properties [57,63].

Material Type	PCM	Density (kg/m ³)		Melting Point (°C)	Latent Heat (kJ/kg)
		Solid	Liquid		
Organic	Paraffin Wax [70]	916	790	64	173.6
	Aluminium [71]	2707	2375	660	397
	Sodium [72]	968	850	97	113
	C ₁₂ H ₂₆ [73]			−10 to −9.3	216
	C ₂₂ H ₄₆ [74]		794	43 to 46	249
	C ₃₀ H ₆₂ [74]		775	64 to 67	252
	C ₈ H ₁₆ O ₂ [75]	981	901	13.3 to 18.4	148
	C ₁₂ H ₂₄ O ₂ [74]		870	42 to 46	171
	C ₁₄ H ₂₈ O ₂ [74]		860	52 to 54	190
	C ₁₆ H ₃₂ O ₂ [75]	989	850	62 to 64	185.4
	C ₁₇ H ₃₄ O ₂ [74]			26 to 29	163.2
	C ₃₈ H ₇₄ O ₄ [74]			63 to 73	215.8
	C ₁₂ H ₂₂ O ₄ [74]			25 to 28	135
	C ₁₉ H ₃₈ O ₃ [74]			49 to 54	126
	C ₂₀ H ₃₈ O ₂ [74]			35 to 37	122
	Ploy Ethylene Glycol [75]			4.2 to 60	117.6
	C ₅ H ₁₂ O ₅ [63]			92 to 96	280
	C ₆ H ₁₄ O ₆ [63]			94 to 96	110
	C ₄ H ₁₀ O ₄ [63]			117 to 119	344
	C ₆ H ₁₄ O ₆ [63]			164 to 169	341
Inorganic	MgCl ₂ ·6H ₂ O [70]	1569	1450	117	168.6
	Mg(NO ₃) ₂ [76]			426	
	Ca(NO ₃) ₂ [77]	2113		560	145
	Na ₂ CO ₃ [77]	2533	1972	854	276
	LiOH [77]	1460		462	873
	NaCl [76]	2160		802	420
	Na ₂ SO ₄ [77]	2680		884	165
	LiF [77]	2640	1810	850	1044
	NaF [77]	2558	1948	995	793
	Zn/Mg(53.7/46.3) [77]	4600		340	185
	Zn/Al(96/4) [77]	6630		381	138
	Al/Si(87.76/12.24) [77]	2540		557	498
	Cu/Zn/Si(74/19/7) [77]	7170		765	125
	Mg/Ca(84/16) [77]	1380		790	272
	Si/Mg(56/44) [77]	1900		946	757
Eutectic Organic	C ₁₂ H ₂₄ O ₂ – C ₁₀ H ₂₀ O ₂ (45/55) [74]			21	143
	C ₁₀ H ₂₀ O ₂ - C ₁₆ H ₃₂ O ₂ (76.5/23.5) [74]			21.8	171.2
	Lauric acid-myristic acid(66/34) [74]			34.2	166.8
	Lauric acid- Palmitic Acid(69/31) [74]			35.2	166.3
	Lauric acid-stearic acid(75.5/24.5) [74]			37	182.7
	Myristic acid- Palmitic Acid (58.0/42.0) [74]			42.6	169.7
	Myristic acid- stearic acid (64.0/36.0) [74]			44.1	182.4
	Myristic acid- stearic acid (65.7/34.3) [74]			52	162
	Palmitic acid-stearic acid (64.2/35.8) [74]			52.3	181.7
Eutectic Inorganic	CaCl ₂ ·6H ₂ O/CaBr ₂ ·6H ₂ O (45/55) [75]			14.7	140
	LiOH/KOH (40/60) [76]			314	341
	KNO ₃ /KCl (95.5/4.5) [76]	2100		320	74
	MgCl ₂ /KCl (39/61) [76]	2110		435	351
	BaCl ₂ /KCl/NaCl (53/28/19) [76]	3020		542	221
	BaCl ₂ /KCl/CaCl ₂ (47/24/29) [76]	2930		551	219
	LiF/MgF ₂ /KF (64/30/6 mol%) [76]			710	782
	LiF/CaF ₂ (80.5/19.5 mol%) [76]			767	790

generally beneficial [80]. The primary challenge with PCMs lies in their poor heat conductivity. However, this issue may be effectively addressed via the use of metallic foam, carbon fibre, and metallic fins inside the container housing the PCM. The use of metallic foam, when restricted to a range of 0.1–3 % of the weight of the PCM, results in a marginal alteration in heat conductivity, but at a substantial expense. Furthermore, it was shown that the incorporation of carbon in the range of 0.1–9 % by weight of the PCM exhibited enhanced performance characteristics [79]. The various thermal conductivity augmentation techniques are described in Fig. 6 and thermal conductivity of PCM with different combinations are illustrated in Fig. 7. The Graphite and other carbon-based compounds are non-toxic materials with a range of desirable features such as high thermal conductivity. However, to improve the thermal conductivity of the PCM, carbon powder has to be added to the PCM. In this investigation, the PCM includes 0.78–2.5 % carbon by volume; larger concentrations were not used as a precaution

to prevent the settling of carbon particles. Fig. 7 shows that, by adding a fin stack to PCM, thermal conductivity may be increased by forty-two times with Al fins and thirty-three times with carbon fins. Planar thermal conductivity of carbon (up to 500 W/m K) is larger than aluminium (~210 W/mK), but the effective thermal conductivity of PCM with Al fins was higher owing to high fin density and uniform fins distribution [82]. In a study the thermo-economic analysis of heating a house with a TES system made of metal foam has been conducted at different percentage of metal foam filling in PCM. The results suggest that the TES unit, when filled to 90 % capacity, shows a substantial decrease in the total time required for complete melting, with a value of 5310 s. This duration is 87.56 % lower compared to a TES unit containing pure PCM [83]. Further study is required for the development of cost-effective and scalable fabrication procedures for PCM composites, particularly those that integrate carbon-based nanoparticles or hybrid architectures. Moreover, the incorporation of multi-functional additives that enhance

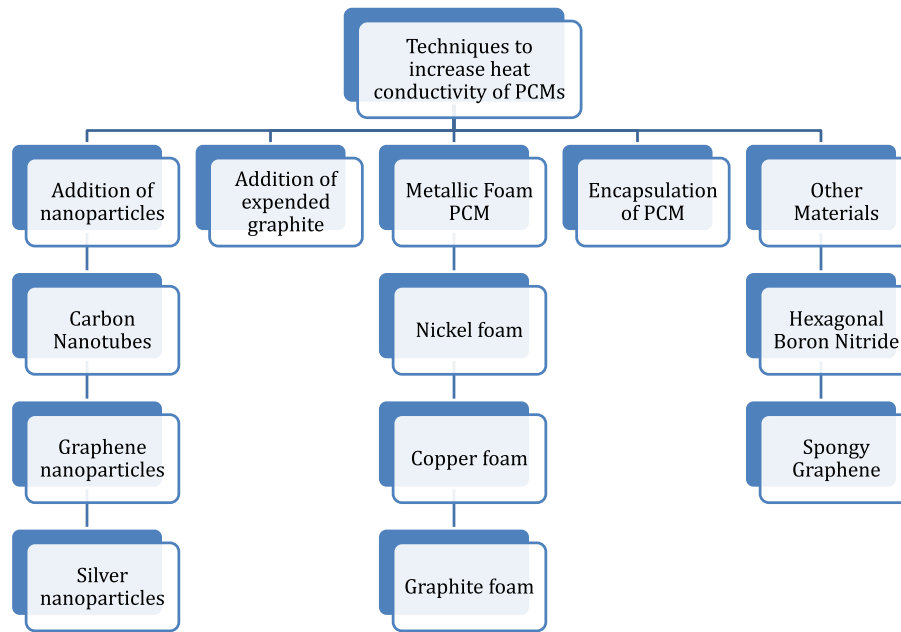


Fig. 6. Thermal conductivity augmentation technique [84].

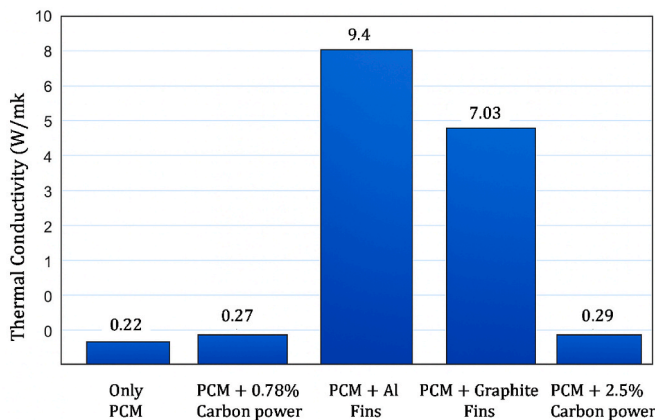


Fig. 7. Thermal conductivity of PCM with different combination [85].

heat conductivity while simultaneously providing form stability or anti-leakage characteristics could yield diverse advantages. Standardized testing methodologies are essential for facilitating reasonable comparisons among various augmentation strategies and enabling material selection for specific TES applications.

3.1.1. PCM nano-composites

PCMs are frequently changed into composite PCMs, so that characteristics of heat transfer can be improved. A PCM composite is made up of a phase change substance, a shape-stabilizing substance (e.g., polymer or ceramic) and a thermally conductive additive (such as graphite). The solidification time is reduced and thermal conductivity could be improved by using this method. Increase in thermal conductivity is correlated with increased nanoparticle ratios. Aside from the positives, they bring with them new hurdles such as increase in cost, compatibility problem, stability issues and increased difficulty [86]. In addition to this PCMs offer an intriguing option for addressing challenges related to TES. In particular, there is a growing interest in nanoparticle-enhanced PCMs [87].

3.1.2. PCM with carbon-based nanoparticles

In a study the fins and MWCNTs have been used to create a scalable PCM-based system with improved heat transmission and light absorption for solar water heating. Experiments with photo-thermal conversion show that a PCM-based system using MWCNTs and fins may produce water temperatures of up to 79.0 °C in a fixed configuration [88]. The research work has been performed using a solar/electro-responsive thermal energy capturing and storage system based on a shape-stabilized heterogeneous porous aerogel composed of polypyrrole (PPy)-deposited CNT and paraffin wax (PW) as the PCM. PW integrated into aerogel supports yields composite PCMs with remarkable phase transition stability after more than 100 heating-cooling cycles and a comparatively high heat storage density of 160.9 J/g. Additionally, by introducing PPy coating as a binding agent in the space between CNTs, the thermal conductivity of the composite PCMs is greatly improved (0.64 W/mK, 2.56 times more than that of PW) [89]. The paraffin wax as PCM with CNT and two supporting material has been taken for the preparation of composite form-stable PCMs with a high photo-thermal conversion function for the purpose of solar TES. The findings indicate that the incorporation of carbon nanotubes (CNT) into the stabilized PCM leads to an enhancement in the efficiency of photo-thermal conversion [90]. In the study, a composite PCM consist of hydrated salt was synthesized and characterized. The impact of addition of HCNTs on the thermal characteristics of the eutectic salt had been investigated using optimization analysis. According to the findings, thermal conductivity of base PCM exhibited a linear rise as the mass percent of HCNTs increased. It reached a maximum value of 1.29 W/(m·K), which was 182.89 % greater compared to the composite without HCNTs and cyclic stability has also been improved [91]. An investigation has been done to create shape stabilized PCM included with MWCNT to enhance thermal latent heat storage capacity as well as heating cooling cycle stability. The new solar thermal storage material exhibited favorable chemical stability during a thermal test consisting of 450 cycles, and had a commendable storage efficiency of around 84 % [15]. The long-term environmental stability in real outdoor direct sunlight conditions (such as moisture, oxidation, and UV exposure) is still not well studied, despite claims of improvements in thermal performance. Additionally, there are still recurring technical issues with the consistent distribution of CNTs within PCM matrices and the scalability of these composite systems for large-scale applications.

3.1.3. PCM with silver nanoparticle

The use of silver nanoparticles into PCMs for solar energy applications aims to enhance both thermal conductivity and overall performance. An experimental investigation was done to examine the impact of silver nanoparticles on improving the thermal energy storage capacity of phase transition materials. Paraffin wax has been chosen as the phase transition material. The mass fraction of silver nanoparticles is reported to be 0.05 % and 0.1 %. The investigation focused on examining the melting and solidification properties of paraffin wax, both in its pure form and when combined with nanoparticles. The research demonstrated that silver nanoparticles in paraffin wax reduce phase transition material melting and solidification temperatures. Increased silver nanoparticle concentration in phase transition materials improves TES. The incorporation of nanoparticles into PCMs is a topic worthy of consideration in order to enhance the performance of solar energy systems [92]. A research investigation was undertaken to examine the thermal characteristics of an organic ester PCM that was infused with silver nanoparticles. The research demonstrates that the incorporation of silver nanoparticles into PCMs may lead to enhanced thermo-physical properties and heat storage capabilities. Consequently, this development holds promise for utilization in solar TES applications [93]. Another recent work, focuses on the experimental investigation of organic PCMs incorporated with silver nanoparticles, with the aim of enhancing their energy storage capacity. The study included the preparation and characterization of organic PCM composites containing silver metal nanoparticles with sizes ranging from 20 to 40 nm. The composites were created at various concentrations, namely 0.2 %, 0.4 %, 0.6 %, 0.8 %, and 1.0 %. The addition of Ag nanoparticles to the base material PCM resulted in a significant enhancement of thermal conductivity. Specifically, the thermal conductivity of the composite increased from 0.2 W/mK to 0.4 W/mK when the mass fraction of nanoparticles was 0.8 %. The increase of 101 % can be attributed to the establishment of highly efficient thermal networks within the PCM matrix. The composite PCM exhibits a latent heat that deviates by 1 % from that of the basic PCM. This discrepancy has significance in terms of enhancing energy storage capabilities, mostly attributed to improved intermolecular attraction. The composites containing scattered Ag nanoparticles exhibited thermal stability up to a temperature of 220 °C, with a small improvement in thermal stability compared to the composites without nanoparticles [94]. An exploration is done to investigate the solar radiation absorptivity and thermal behavior of salt hydrate PCM mixed with silver nanoparticles. The results indicate that the composite salt hydrate sample with a greater concentration of silver nanoparticles will have improved optical and thermal behavior [16]. It is important for nanoparticles to be evenly spread out without clumping together in order to maintain performance, but many studies do not pay enough attention to this. Silver is expensive, and the effects on the environment of nanoparticles leaking out or breaking down over time have not been thoroughly studied. Also, there isn't any long-term cyclic thermal stability data, especially not under real solar field conditions.

3.1.4. PCM with expanded graphite

A study has been conducted on effect of expended graphite with PCM on heat transfer characteristics of solar water heating system. The experimental findings indicated that the rate of volume expansion for a 16 L EGPHE was determined to be 6.25 % when subjected to a temperature of 76 °C. A comprehensive comparison was conducted to examine the disparities in hot water flow between the EGPHE and a conventional water tank. For supplied water temperatures of 45 °C and 40 °C, the volume of heated water increased by 83.12 % and 194 %, respectively, when the input flow rate was 0.5 L/min. When the rate of flow at the intake was raised, there was a corresponding drop in the rate at which the volume of hot water rose. The efficiency of the EGPHE system can be enhanced by the reduction of exergy destruction [95]. Nonadecane PCM embedded with AC-EG (Activated carbon-Expanded graphite) hybrid structure to address the issue of leakage and poor

thermal conductivity of storage material. The utilization of AC-EG hybrids as carrier matrix structures and thermal conductivity enhancers for Nonadecane PCM, along with the production of various leak-proof composite PCMs, holds significant potential for enhancing low-temperature thermal management systems in applications such as envelopes for buildings, electronic devices/equipment, food transportation, and textile products [83]. There have not been a lot of researches done on how different porosity and EG structure affect the thermal performance of different PCM materials. There should be more research on how to optimize the size and distribution of EG particles in the PCM matrix, how the porosity of the composite affects the speed of phase change, and how well these materials can be scaled up for use in commercial energy systems.

3.1.5. PCM with metal foam

A study investigates the melting and solidification behavior of PCM inside a horizontal tank that is filled with graded metal foam. The investigation employs both experimental and computational approaches to get a comprehensive understanding of the process. The findings shows that foam with PCM improves the heat storage capacity of system [96]. The solidification and melting of PCM encased in a porous metallic foam structure within a horizontal multi-tube thermal storage system was investigated in a research project. Through the use of the MTHX programme, a simulation of the melting and solidification process of PCM embedded metal foam was carried out. The use of metallic foam and the augmentation of inner tubes may enhance the efficiency of the phase change process. Utilizing metallic foam in solidification demonstrates more efficacy than its usage in melting. The efficiency of tube arrangement inside a composite PCM is higher compared to that within pure PCM [97]. These studies show that metal foam is a good way to improve heat transfer during phase changes, especially when things are solidifying. But the effect on melting is very small, so it is necessary to learn more about how the structure of foam and temperature differences are connected. Even though horizontal layouts are often looked at, there is still disagreement about the best orientation, porosity, and foam density for some types of PCM and thermal loads.

3.1.6. Copper porous material

An evacuated solar collector based on heat pipes and including a PCM incorporated into copper porous material has been used in solar water heating systems. Furthermore, a comparative performance study has been conducted, contrasting this innovative collector with a traditional solar collector. The results showed that the evacuated tube with PCM + Cu porous metal saw a significant increase in maximum temperature, reaching around 21 °C, at the solar radiation maximum times. After sunset, a 36.1 °C temperature differential was noted between the evacuated tube containing PCM + Cu porous metal and the regular evacuated tube. Furthermore, it was observed that there was a maximum temperature differential in the water output of 8 °C, 10 °C, and 11 °C between the two systems during the time period of 10 p.m. on April 9th to 11th, respectively. The recommended system's greatest energy efficiency was 85.64 %, compared to 36.91 % for the old system's maximum daily energy efficiency [98]. The copper-based metal foam with paraffin PCM has been used to improve the performance latent heat storage system. The result shows that the PCM with metal foam reduces the melting time during heat storage [99]. In an experiment a vacuum impregnation process is employed to immerse copper foam and nickel in paraffin. The PCM impregnation ratios in composite PCM with foam-based nickel can range from 96.0 to 98.6 %. The thermal conductivity of the composite material based on paraffin and nickel foam was increased by a factor of three, while the phase change content of the composite material based on copper foam was increased by a factor of fifteen [100]. A recent study has been conducted on utilization of anisotropic metal foam layer for enhancement in heat storage capacity of PCM in solar water heating system. The study suggest that strategic placement of anisotropic layer of metal foam improves the thermal

efficiency of system and also it reduces the PCM melting time by 13.12 % [101]. Despite stated performance improvements, there are few quantitative comparisons between copper-based materials' various pore sizes, foam densities, and geometries, which makes it challenging to generalize the best design practices.

3.1.7. Aluminum foam porous material

A study was conducted to improve the heat transfer capabilities of paraffin, used as a PCM in solar flat-plate collector systems for domestic and industrial solar applications. This was achieved by incorporating aluminium foams with porosity of 0.92 or 0.95, adding different types of nanoparticles at a percentage of weight of 5 %, and modifying the fin geometry in three different configurations. This implies that metal foams have a significant impact on enhancing cycle durations in these particular contexts. The use of aluminum metal foam and nanoparticles results in a significant decrease of 87.21 % in charging times [102]. One potential approach for addressing the issue of lower thermal conductivity of PCM is to integrate the PCM with a highly conductive substance such as metal foam and/or nanoparticles. In a study the TES system under consideration is a standard 70 L water tank that has been filled with nano-PCM. The system incorporates pipes to facilitate the transmission of thermal energy from a fluid medium to the nano-PCM material. The PCM used in this study is composed only of paraffin wax, whereas the nanoparticles employed consist of aluminium oxide. The metallic foam is composed of aluminium and has certain porosity values. The use of the enthalpy-porosity concept is applied in order to simulate the phase transition of the nano-PCM, while the metal foam is represented as a porous medium inside the model [103]. From the above discussion, it has been found that the development of hybrid foam structures (e.g., graded porosity or composite foams) that provide customized heat conductivity and mechanical strength needs more exploration.

3.1.8. Boron nitride embedded PCM

Researchers have investigated the integration of boron nitride (BN) or hexagonal boron nitride (h-BN) with PCM to enhance its thermal energy storage capacities. Hexagonal boron nitride (h-BN) is a 2D material known for its remarkable thermal conductivity and electrical insulating properties. These qualities render it a very attractive contender for integration into electronic systems, particularly for the purpose of thermal control. In this view, an investigation involves the experimental evaluation of a newly developed hexagonal boron nitride nanocomposite, which is based on a eutectic binary molten salt. Four distinct kinds of samples were generated, which consisted of pure binary molten salt and binary molten salt-based h-BN nanocomposite. The loading concentrations of h-BN in the nanocomposite were 0.5, 1, and 1.5 wt%. The addition of h-BN nanoparticles to binary molten salt improved its thermal stability, as shown by the results. A 1.5 wt% loading concentration of h-BN nanocomposite in a binary molten salt showed an improvement in thermal stability of 16 % [104]. According to another a research study, the current thermal conductivity of PCMs is quite low, often measuring less than 1 W/mK. This limited thermal conductivity significantly hampers the ability of PCMs to effectively dissipate heat in thermal applications. The study focuses on the development of a composite material including hexagonal boron nitride nanosheets (BNNs) and commercially available paraffin wax. The BNNs provide desired properties including electrical insulation and high thermal conductivity to this composite material, while the paraffin wax contributes a substantial heat of fusion. The thermal conductivity of composites made up of paraffin wax and BNNs is increased to 3.47 W/mK by the use of BNNs. This represents a twelvefold increase in comparison to the thermal conductivity of pure wax, which is 0.29 W/mK [105]. A recent research has been undertaken to investigate the enhancement of thermal properties in three distinct organic PCMs inserted with hexagonal boron nitride (h-BN). The results of the study indicate that the addition of 2 wt% (h-BN) is sufficient to provide a good boost in thermal conductivity (TC) without adversely impacting the

latent heat storage (LHS) capabilities of the chosen PCMs [106]. Therefore, hexagonal boron nitride (h-BN) reinforced PCMs have the potential to be used in the solar collectors, solar photovoltaic (PV) systems, and home water heating systems, among other applications [107]. An experiment was conducted to investigate the potential of hexagonal boron nitride as a thermally conducting infill in organic PCMs. Incorporating h-BN into pure paraffin PCM leads to a substantial enhancement in heat transfer characteristics. This results in a remarkable 192.2 % rise in thermal conductivity and a notable 204.1 % increase in thermal diffusivity [106]. The enduring reliability of BN-embedded phase change materials under actual operational conditions requires more research in addition to the study of the impact of particle alignment and interfacial adhesion on thermal conductivity.

3.1.9. Spongy graphene embedded PCM

The composite of graphene sponge embedded with PCM has great potential as an energy storage material. A research examines the use of a composite material comprising of 3D graphene sponge and PCM to improve the thermal efficiency of thermal management systems (TMS). This work presents the development of a TMS with high integrity. The TMS is constructed using a combination of paraffin wax, reduced graphene oxide, and melamine sponge, which is a 3D graphene sponge. This presents a novel design for a PCM system that offers a complete range of performance attributes, including lightweight construction, effective temperature management capabilities for low as well as high temperatures, and reliable leak-proof functionality [108]. Graphene sponge PCM composite is another potential energy storage composite PCM that might be used in TES and TMS.

3.2. Enhancement using PCM encapsulation

Encapsulation, which encases the PCM into a shell material with higher value of thermal conductivity, is also a way to improve the value of thermal conductivity of PCMs (e.g. silica). Furthermore, this strategy increases the heat transfer surface while increasing the amount of accommodation. Encapsulation is an effective way to increase heat conductivity [85]. Capsules may exhibit three distinct size ranges, namely macro- (>1000 μm), micro- (1–1000 μm), or nanosized (1–1000 nm). The use of smaller capsules significantly enhances the ratio of surface area to volume of the material, hence leading to an improvement in heat transmission [109]. The use of slurries and dispersions containing micro or nano-encapsulated phase-change materials (ePCMs) has the potential to significantly transform the solar thermal industry. This is because these materials possess the unique ability to both absorb solar photo thermal energy and store it [110]. However, this is the most difficult and expensive of the three approaches provided here [86]. Micro-encapsulated CaCO_3 shells with n-octadecane PCM is used to improve heat conductivity. According to the study, the thermal conductivity of encapsulated n-octadecane increased from 0.153 W/m-K in pure PCM to 1.264 W/m-K [111]. Another researcher did similar work with a different PCM (melting temperature of 26.57 °C, paraffin RT-28 PCM). The microcapsules paraffin CaCO_3 , conductivity improved from 0.289 W/m-K in pure PCM to 0.759 W/m-K [112]. A conducted work focuses on the microencapsulation of beeswax, a bioderived PCM, using an ethyl cellulose shell. The objective is to investigate the potential of this encapsulated material for TES applications. The microcapsules that have been synthesized consist of sustainable components and have a high TES capability [113]. The thermal study demonstrates that the process of microencapsulation of PCM improves the efficiency of TES [114]. The use of PCMs with nano encapsulation has gained significant interest due to its prominent features such as high specific surface area, outstanding suspension stability, and favorable structures for energy management and storage. The most common methods of nano encapsulation is illustrated in Fig. 8. It has been found that the interfacial thermal resistance between core and shell materials, especially at the nanoscale, remains inadequately comprehended and optimized. Also, limited

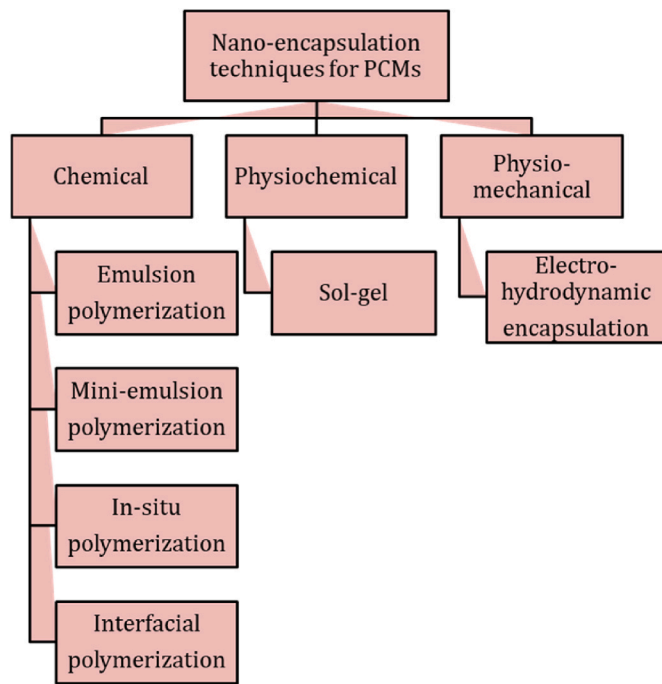


Fig. 8. Nano encapsulation technique for PCM [53].

research examines the environmental effects and disposal methods of nanoencapsulated phase change material composites at the end of their life cycle.

3.2.1. Microencapsulation of PCM

PCMs are increasingly gaining popularity and recognition in the field of TES. Microcapsules enhance the mechanical and thermal properties of PCMs used for TES by increasing the heat transfer area and minimising the seepage of melted materials. The use of PCM encapsulation is necessary in order to prevent leakage of PCM, enhance heat transfer, and effectively manage and minimise the reactivity of PCM with the surrounding environment. In recent times, a considerable body of study has emerged about the use of PCM microcapsules in energy systems, with several studies being conducted to elucidate their practical applications [115]. The examination focuses on the utilization of microencapsulation techniques used to bioderived PCM beeswax, with an ethyl cellulose shell, for the purpose of TES applications. Beeswax is a naturally occurring PCM that has a significant phase transition enthalpy, which is equivalent to that of synthesized PCMs. The emulsion solvent evaporation technique was used to shape-stabilize beeswax using the biopolymer ethyl cellulose. The microcapsules that have been synthesized consist of sustainable materials and have a significant TES capability. The synthetic microcapsules have the potential to serve for TES applications [113]. To improve TES capacity, researchers have recently investigated microencapsulating bio-based PCM with an inorganic shell covered with silica. The cenospheres is used for the encapsulation of a cost-effective, bio-derived PCM obtained from processed edible vegetable oil. The results of this study support the idea that cenospheres might serve as an inorganic covering for PCMs and that they could one day be used into construction materials for TES [116]. The development of scalable and cost-effective encapsulation systems that make use of eco-friendly shell materials requires additional research. It's possible that hybrid shells, which combine organic and inorganic components, could provide a more equal approach. In addition, lifecycle analysis, mechanical performance, and integration with host materials (such as cement and polymers) are to be focused for future applications of TES.

3.2.2. Nanoencapsulation of PCM

The limited thermal stability, poor thermal conductivity, and liquid nature of PCMs provide obstacles in direct application. The PCM may be shielded from outside effects, given more surface area for heat transmission, and made possible by encasing them in shell material, which also enables mobile thermal energy storage devices [117]. Nano-encapsulation has emerged as a very promising approach for enhancing the efficacy of PCMs, including both organic and inorganic variants. The aforementioned phenomenon facilitates the augmentation of a substantial specific surface area, impedes the interchange of the enclosed substance with the surrounding environment, regulates thermal transfer via the capsule membrane, and triggers corresponding melting or crystallisation owing to the diminutive size of the core [109]. An investigation focuses of on the preparation of nano-encapsulated PCM emulsions using a solvent-assisted approach, with the aim of exploring their potential applications in solar energy systems for efficient energy storage with better PCM stability [109]. Researchers have done examination on the preparation and characterization of nano-encapsulated n-tetradecane, which serves as a PCM for TES. The use of nano-capsules has the potential to be employed in the context of TES and the increase of heat transport [118,119]. It is evident that, choosing the right encapsulation size is essential for achieving efficient and long-lasting thermal energy storage in solar water heating systems. Fig. 9 illustrates the three main types of PCM encapsulation based on capsule size: nano, micro, and macro-encapsulation. Nano-encapsulation (less than 1 μm) allows for quick heat transfer due to its high surface area, while micro-encapsulation (1–1000 μm) offers a good balance between thermal performance and structural stability. Macro-encapsulation (greater than 1 mm) holds larger amounts of PCM but tends to have slower heat exchange rates. In each case, the protective shell helps maintain the integrity of the PCM and prevents leakage.

Electrohydrodynamic processing is one exclusive technique utilised for PCM material encapsulation. This type of processing has shown to be effective for encapsulating a variety of components, including PCMs, biomedical compounds, functional food additives, and other chemicals, in polymer matrices [117]. Another technique using nano-encapsulated PCM for improved performance is the sol-gel approach. In this regard a study focuses on the synthesis, characterization, and thermal characteristics of nanoencapsulated phase transition materials using the sol-gel technique. The PCM nanocapsules consist of a core material composed of palmitic acid (PA) and a shell material composed of SiO_2 . The PA/ SiO_2 nanocapsules that have been developed demonstrate suitability as PCMs for slurry TES applications due to their favorable thermal characteristics, reliable thermal performance, chemical stability, homogeneous shape, and thermal conductivities [81]. However, various macro and micro encapsulation materials for PCM have been shown in Table 3. Nanocapsules in real-world TES systems have not yet been thoroughly investigated with regard to their long-term influence on the environment and their degrading behavior. Also, it is not possible to obtain sufficient comparative data on the economic and energy costs of various nanoencapsulation techniques when applied on a large scale. Research has not yet been conducted to determine whether nano-encapsulated PCMs are compatible with complicated system topologies, such as those found in solar thermal collectors or building-integrated thermal energy systems (TES).

4. PCM encapsulation technique in solar water heater

In the present study, some recent solar water heater designs that uses PCM as the TES material has been reviewed. The PCMs can be used with solar water heater in the following components:

- i. Water storage tank
- ii. Solar collector
- iii. Insulated pipework.

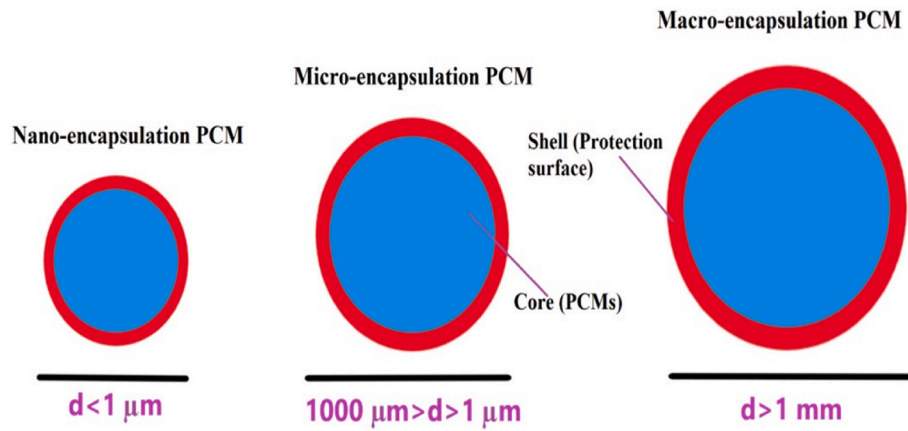


Fig. 9. PCM encapsulation technique based on the size of PCM [39].

Table 3
Encapsulation materials for PCMs.

Encapsulation Material	Material of Shell	Material of Core	Thermal Stability (°C)
Macroencapsulation	Steel	Paraffin Wax	36 [120]
	Cr-Ni	Cu	>1150 [121]
	Stainless steel 304 L	MgCl ₂	>750 [122]
	SiO ₂	NaNO ₃	300-500 [123]
	Ni	Pb	400 [124]
Microencapsulation	Ni	Cu	>1100 [125]
	TiO ₂	Paraffin Wax	240 [126]
	Na ₂ SiO ₃	n-Octadecane	>375 [127]
	M-F	C ₁₆ H ₃₄	388 [128]
	VTMS	n-Octadecane	318.2 [129]
	CaCO ₃	n-Octadecane	215 [111]
	C	Palmitic acid	180-320 [130]
	SiO ₂ paraffin	Paraffin Wax	292 [111]
	Al(OH) ₃	Palmitic acid	200-520 [131]

4.1. PCM encapsulation in storage tank

The PCM was commonly integrated into the SWH storage tank during the early years of research and development. An experiment has been conducted on development and testing on PCM filled at the bottom of a storage tank in 1985 [132]. PCMs are used in solar water heater storage tank to store solar energy for a longer period of time and to use it when solar radiation isn't accessible. PCM can be incorporated in many ways in storage tank. As shown in Fig. 10(a), the PCM is installed in such a manner that a single circular block is introduced in a storage tank.

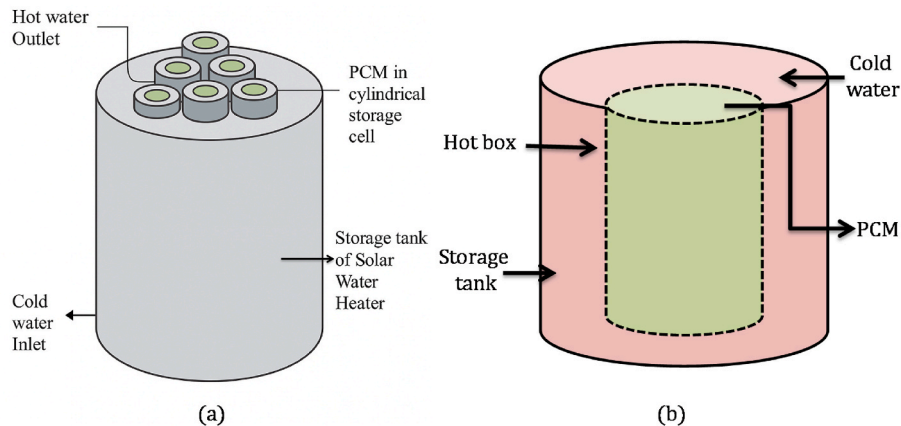


Fig. 10. Illustration of PCM in (a) Circular cylindrical block and (b) Array form in a storage tank.

PCM are also arranged in an array form as shown in Fig. 10(b). Electro hydrodynamic processing is one exclusive technique utilised for PCM material encapsulation [117]. Many PCM of cylindrical form are employed in storage tank. It increases the heat storing capacity. This research investigates PCM thermal storage in traditional water tank systems. Traditional sensible TES tanks and hybrid latent heat storage tanks are compared. In the hybrid system, cylindrical nodules encapsulate the PCM and incorporate it into the water tank to increase its energy density. According to the experimental results, adding PCM to a water storage tank increases its capacity for heat storage by 15 % when compared to a conventional tank that just holds water [133]. An experiment was conducted to explore the integration of paraffin PCM into the storage tank of a SWH, with the aim of achieving a constant output temperature for hot water. The use of an encapsulated paraffin wax PCM in a specific solar water heating system has shown the capability to provide a volume of 1200 L of heated water, maintaining a temperature range of $60\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ for a duration exceeding 8 h [134]. A research aims to assess the efficacy of incorporating PCMs into a warm water storage tank in a SWH system, using theoretical simulation techniques. The Sodium hydrogen phosphate dodecahydrate (SHPD) has identified the greatest efficacy factor value of 3.27 among all PCMs (PCMs). As a result, using this specific PCM for the SWH system's storage tank is strongly recommended. The study also examines the melt percentage of several PCMs with respect to materials used as containers, including copper, aluminium, glass, stainless steel, and aluminium and tin. Based on the investigation conducted, it has been shown that copper, as a container material, has a much higher melting rate for all PCMs in comparison to other container materials. Consequently, copper may be regarded as superior in terms of its performance in this regard [135].

The use of cylindrical capsules for encapsulating PCM has shown significant potential in enhancing the heat transfer properties inside LHS systems. The investigation of heat transmission characteristics inside the PCM is conducted by the graphical representation of temperature variations at various positions within the capsules. The experiment recorded that the LHS module, including about 14 kg of PCM, exhibited a total heat storage/discharge of 3.97/3.94 MJ within the specified temperature range [136]. More research is required on standardized testing, lifespan analysis of encapsulated PCM modules, and hybrid systems that have both latent and sensible storage for the best results. Furthermore, investigating PCM behavior under fluctuating load requirements and creating predictive models for system performance would facilitate the broader implementation of these technologies in practical solar thermal systems.

4.2. PCM encapsulation in solar collectors

In 2014, research has been conducted to design and build a solar collector with flat plate integrated with PCM, as depicted in Fig. 11. The upper surface of absorber plate is covered with nine riser tubes, while the absorber plate's bottom surface was covered with 37 fins. PCM was put into the absorber plate's bottom. The fins attached to the rise absorber tube worked as heat exchanger unit. Fin increases heat transfer rate between absorber plate and PCM by increasing surface area of heat transfer. In this study, PCM is employed in two ways: i) paraffin wax; ii) paraffin wax nanocomposite containing 20 nm Cu-nano particles with 1.0 wt%. Three studies: (i) without PCM, (ii) with PCM, and (iii) and with the Cu-PCM nanocomposite, were tested at different inclination angle like 10°, 20° and 30°. The water tank temperature for the system without PCM was found to be 35.1 °C at 7.00 a.m. after 24 h of operation. The water tank temperatures, on the other hand, were 40.7 °C and 40.1 °C for the systems with Cu-PCM nano-composite and PCM, respectively. The efficiency of solar collector has been found to be increased by 6.9 % for paraffin wax and 8.4 % for Cu-PCM nano-composite [137].

An endeavour is undertaken to examine and determine the optimal arrangement for integrating LHTES into an ETCSWH. The research employs the use of ETCSWH integrated with PCM in three different configurations, while keeping the operating factors same. The findings indicate an inverse relationship between the cylinder number and the period necessary for full melting. Moreover, the placement of containers within the ETCSWH system is a crucial determinant in assessing the potential advantages of incorporating LHTES [138]. The impact of using a unique PCM encapsulated design on the thermal performance of a solar collector has been studied. A slender stratum of PCM was strategically placed underneath the absorber plate, serving as a rear layer to store the accumulated latent heat and the impact of PCM on the

temperature of the water at the exit was investigated. The results demonstrate a high level of satisfaction and acceptability for extended periods of use in both residential and industrial settings [139]. From the above study, it has been found that the comparative performance of different PCM materials and encapsulation geometries under dynamic weather conditions is still inadequately investigated. Also, Investigation of sustainable and bio-derived phase change material options with enhanced thermal and ecological efficacy need more exploration.

4.3. Insulated pipework with PCM

In a study, a design has been created for shell and tube SWH, as depicted in Fig. 12. In-between the annular space of outer shell and copper tube, paraffin wax as PCM is used having melting point 56.06 °C. Thermal solar energy storage unit is attached to thermal bath. A double pipe thermal storage, solar collector and an insulated storage tank of water, are the three major components of the system. Energy efficiencies were found to be 63.8 percent and 77.4 percent, respectively, when using water at flow rate of 1.98 kg/h and 10.02 kg/h, although the exergy analysis revealed that efficiencies were around 9.58 percent and 6.02 percent [140].

5. Enhancement methods of PCM-based TES

In recent years, researchers have been actively working to improve the performance of PCM-based thermal energy storage (TES) systems used in solar water heating. One of the main challenges with PCMs is their low thermal conductivity, which limits how quickly they can absorb and release heat. To address this, several enhancement methods have been developed. A common strategy is to mix the PCM with materials that conduct heat better such as metal fins, porous metal foams, or nanoparticles like copper oxide, aluminum oxide, or graphene. These additives help the PCM melt and solidify more quickly, making the storage system more efficient. For example, studies in top-tier journals like Renewable Energy and Applied Thermal Engineering have shown that adding graphene or copper nanoparticles to paraffin wax can speed up the heat transfer process by 30 %–50 % [134]. Another approach focuses on improving the design of the storage container. By changing the shape or adding internal structures like fins or layers, the system can transfer heat more effectively. There's also growing interest in using composite PCMs by combining different materials or creating eutectic mixtures, researchers can fine-tune the thermal properties for better performance. Encapsulation techniques, such as micro and macro-encapsulation, are also being used to prevent leakage and ensure long-term stability. These combined efforts are making PCM-based TES systems more practical, efficient, and suitable for everyday use in solar

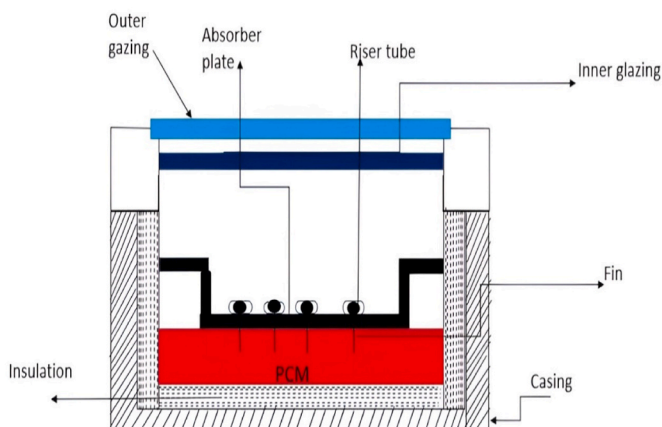


Fig. 11. Solar water heater collector with PCM [137].

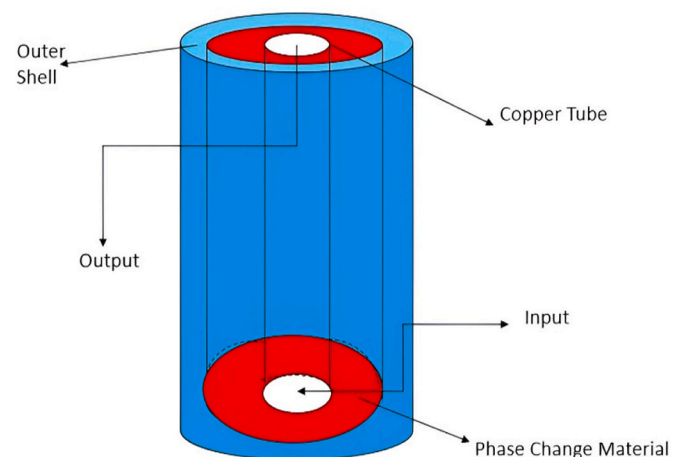


Fig. 12. Insulated pipe work with PCM [140].

water heating applications.

5.1. Fin based heating system with PCM

Recently a novel idea is proposed for in line solar water heater in which asymmetrical compound parabolic collector has been combined with PCM. The solar collector is designed on thermal as well optical basis. The optical performance has been optimized with two independent reflector design. Furthermore, four different kind of internal fin design have been compared and optimized. The fin based system gives substantial improvement in system performance [141]. Another study is conducted on FPSWH with modified riser design. The existing riser tube has been integrated with semicircular riser tube with fin and PCM. The modified riser design heightened the thermal performance of PCM and 177.7 % rise in heat transfer and 39.5 % rise in efficiency have been recorded [142]. A study is conducted on comparison of performance of modified compact fin based solar water heater with two PCM for domestic single family use. The new proposal proposes a hybrid and modular storage component with multiple cavities holding each the PCM, delimited by two flat heat exchangers joined by slotted fins. The distance between the fins placed, and thermal conductivity of PCM plays an important role in heat transfer [14]. These studies show that using fins with PCM in solar thermal systems can improve performance in different ways, but they mostly focus on experimental validation and don't look at long-term operational stability, cost implications, or scalability. Most designs improve heat transfer by increasing the surface area of the fins or arranging them in the best way. However, they don't look at the trade-off between higher material costs and thermal benefits in real-world situations.

5.2. Modified evacuated tube collector

Modified evacuated tube collectors (ETCs) have emerged as a highly efficient solution for integrating PCMs into SWH systems. Traditional ETCs are known for their superior heat retention due to vacuum insulation, but recent modifications have aimed to further enhance their thermal performance and storage capabilities. These improvements typically involve structural changes such as incorporating finned inner tubes, U-tube configurations, or concentric tube designs to increase the surface area for heat transfer. More advanced designs embed PCM capsules or jackets within or around the inner tube, allowing for direct heat absorption and extended heat release even after sunset. Literatures show that such modifications can significantly boost the thermal efficiency of ETCs, in some cases achieving up to 20–30 % more effective heat retention compared to unmodified designs [145]. A modified design of evacuated tube collector has been used for domestic solar water heater. Myristic acid is taken as heat storage material. The main benefit of this design is that the incorporation of myristic acid in the manifold does not impact the vacuum tube's strength and does not take up extra roof space. Myristic acid's excellent thermal and chemical characteristics increase system performance. The higher and lower flow rates during both charging procedures are advised for high daily efficiency and high hot water temperature, respectively. The modified design meets the hot water demand in the late evening and reduces the risk of overheating as compared to the standard one [144]. Overall, modified ETCs provide a more reliable and energy-efficient option for solar water heating, especially in regions with fluctuating solar radiation or high evening water demand.

5.3. PCM with hybrid nano particle

One of the promising recent advancements in PCM-based thermal energy storage systems is the incorporation of hybrid nanoparticles to enhance thermal conductivity and overall performance. Unlike single nanoparticles, hybrid nanomaterials combine two or more types of nanoparticles such as metal oxides and carbon-based to achieve

synergistic effects. These hybrid nanofillers improve not only the thermal conductivity but also the heat distribution within the PCM, leading to faster charging and discharging cycles. Studies have shown that hybrid-enhanced PCMs can offer up to 60–80 % improvement in thermal conductivity compared to pure PCMs. In a recently conducted study, the concept of hybrid nanoparticle based PCM in FPSWH has been examined. The hybrid nano PCM (paraffin wax) has been developed using various weight percentages of MWCNTs and nano SiO₂. Paraffin wax's thermal conductivity was 0.24 W/m-K, but for hybrid nano PCM, it rose to 0.47719 W/m-K. Collector efficiency with hybrid nano PCM is found to be 71.7 % [145]. Moreover, the hybrid approach often requires a lower concentration of nanoparticles, reducing the risk of agglomeration and maintaining the latent heat capacity of the PCM. These findings suggest that hybrid nanomaterials offer a balanced and efficient way to overcome the limitations of conventional PCMs, making them a promising solution for next-generation solar thermal systems.

6. Melting and solidification characteristics of PCMs

The radiation falling on the solar devices are converted into heat and the heat can be stored in PCMs. TES in PCM, is the effective way to store thermal energy. However, PCMs suffers with many problems like, lower thermal conductivity, lower stability, and leakage during melting and large degree of supercooling. Higher degree supercooling of PCM affect the system performance negatively during charging and discharging of energy from PCM. In solar thermal applications, the PCM should have lower degree of supercooling. This indicates that there should be little variation between the melting and solidification points. In order to make PCMs more suitable for solar thermal applications, several researchers have sought to lower their supercooling degrees. Safari et al., [146] have explored that PCMs have a thermo physical feature known as supercooling, which creates problem during use of PCM in solar thermal energy storage. Zahir et al., [147] have mentioned that the issue of supercooling restrict the applicability of PCMs in solar thermal applications. The use of nucleating agents, thickeners, and macroporous structures are the most prevalent ways to mitigate the consequences of supercooling. These agents include carbon nanotubes, fine salt particles, nano additions, and carboxy methyl cellulose. Kant et al., [148] have investigated the solidification and melting behavior of PCMs during cyclic melting and solidification. The melting time of PCM is noted to be less than the solidification time, and it is also noted that the melting time is shorter with changing heat flux than with constant heat flux. Abdulateef et al., [149] have numerically analyzed the Triplex-Tube Heat Exchanger (TTHX) using paraffin (RT82) and alumina nanoparticles (Al₂O₃) with solidification and melting temperature of 65 °C and 82 °C, respectively. According to the results, the liquid fraction and isothermal PCM contours produced employing external triangular fins were fully achieved at 193 min and 630 min, respectively. A recent research investigates the enhancement of melting and solidification performance of PCM by the use of a unique branch fin structure. A unique fin structure was introduced with the aim of enhancing the heat transmission properties of a triple-tube latent heat storage system. It was noted that the inclusion of fins resulted in a reduction of 67.7 % and 74.8 % in the melting and solidification time, respectively, as compared to the scenario where no fins were present. Under optimized conditions, the duration needed to complete a cycle of heat storage and release is decreased by 84.5 % in comparison to a scenario where fins are not included [150]. Huang et al., [151] have examined the solidification and melting induced by buoyancy in finned LHS. The findings showed that for gallium and n-octadecane, the buoyancy-driven melting durations increased by 1.8 % and decreased by 32.6 %, respectively, in comparison to conduction-dominated situations, whereas the corresponding solidification periods decreased by 8.4 % and increased by 15.2 %. Recently an investigation is done to examine the impact of an intelligent memory metal fin on the melting and solidification process of PCM via experimental and computational methods. The experimental findings

demonstrate a decrease of 28.6 % in the total melting duration of the PCM while employing the intelligent memory metal fin, as opposed to utilizing the straight fin. The simulation findings indicate that there is a reduction in the melting and solidification time as the length of the intelligent memory metal fin increases. They introduced a novel approach to improve the melting and solidification process of PCM in a square cavity TES unit by using an intelligent memory metal fin [152]. Despite a lot of research, there is still no standard way to evaluate PCM performance. Also, modified phase change materials, especially those with nanoparticles or complex shapes, need to be studied more thoroughly to find out how stable they are over time and in terms of temperature and structure. It hasn't been fully proven that new fin structures and advanced materials can be used in real solar thermal systems. Development of smart PCM systems that can change its behavior on their own (for example, by using smart fins) to account for changes in solar input.

The radiation falling on the solar devices are converted into heat and the heat can be stored in PCMs. TES in PCM, is the effective way to store thermal energy. However, PCMs suffers with many problems like, lower thermal conductivity, lower stability, and leakage during melting and large degree of supercooling. Higher degree supercooling of PCM affects the system performance negatively during charging and discharging of energy from PCM. In solar thermal applications, the PCM should have lower degree of supercooling. This indicates that there should be little variation between the melting and solidification points. In order to make PCMs more suitable for solar thermal applications, several researchers have sought to lower their supercooling degrees. During phase transitions in phase change materials (PCMs), Yadav et al. [153] investigated ways to improve thermal conductivity, lower thermal instability, and stop leakage. EG particles (20 wt percent) were infused into liquid A70 (PCM) to create new shape-stable phase change materials (SS-PCMs). With a remarkable thermal conductivity of 1.59 W/(m·K), it outperformed base (A70) PCM by 657.16 %. The SS-PCM composites were also subjected to 500 thermal cycles, and steady wavenumbers before and after the 500 cycles show thermal stability. The absence of new peaks suggests that there have been no chemical reactions or heat damage. By dispersing functionalized multi-walled carbon FMWCNT in weight concentrations of 0.2, 0.4, 0.6, and 0.8 % in the presence of stabilizer SDBS, Rajamony et al. [154] improved the thermophysical properties of salt hydrate (SH) PCM. Its thermal conductivity increased by 91.45 % at 0.6 wt percent MWCNT. Also, after 300 cycles, nano-composites showed thermal and chemical stability. Salt hydrate PCMs make the solar heating system more efficient by decreasing fluctuations in temperature and reducing the requirement for constant solar input. Additionally, Rajamony et al. [155] investigated on nano-enhanced salt hydrate PCM based on binary carbon (graphene and multi-walled carbon nanotubes). In comparison with pure salt hydrate, it was found that adding 0.07 wt percent MWCNT and 0.7 wt percent graphene to salt hydrate significantly increased its thermal conductivity by 160 % (1.2 W/mK). Furthermore, even after 300 thermal cycles, the PCM composite improved by binary nanoparticles remains chemically and thermally stable up to 473 °C. Moreover, after 300 heating cycles, the binary nanocomposite's latent heat is nearly identical. Rajamony et al. [156] conducted research on the synthesis of environmentally friendly coconut shell biochar (CSB) in varying weight concentrations (0.2 % and 0.8 %) that were dispersed with organic A46 PCM. In comparison to basic PCM (0.22 W/m K), the produced 0.8 % biochar concentration composites exhibit higher thermal conductivity (0.39 W/m K). Additionally, composites typically demonstrated strong chemical stability, suggesting long-term application possibilities.

7. Technical challenges of PCM/NePCM integrated solar water heater

Organic PCMs have a low thermal conductivity, which indicates that the device is not very efficient because of inefficient charging and

discharging times [157]. In order to improve the thermophysical characteristics of PCM, different nanoparticles are added [158]. It has been found that while this improves thermal conductivity, it also presents problems, including higher viscosity and the possibility of nanoparticle fusion. Additionally, it is discovered that when thermal cycling procedures are repeated, solar water heaters based on PCM/NePCM do not exhibit stability. Over time, repeated cycles of melting and solidification may cause PCM characteristics to deteriorate, which could impair the thermal storage system's long-term performance. Degradation over time may result from certain PCMs' corrosive or chemically reactive properties with typical building materials [159]. Some PCMs, particularly inorganic types such as salt hydrates, are susceptible to phase separation and supercooling, which is the tendency to remain liquid below the freezing point when subjected to thermal cycling [160]. These occurrences may result in less effective energy storage and uneven thermal performance. Phase transitions also cause volume changes in PCMs, which can put stress on containment structures. Solar water heater tanks and storage tubes may lose their structural integrity due to frequent expansion and contraction. Certain PCMs are poisonous or flammable, which makes their use dangerous. Selecting safe and non-toxic PCMs, such as some paraffins, is essential for household hot water applications to avoid health risks in the event of leaks or system malfunctions. The design of solar water heaters becomes more complex when PCMs are used, necessitating the installation of other parts like control systems or heat exchangers. Technical Challenges of PCM/NePCM-Integrated Solar Water Heaters as shown in Infographic Fig. 13. In order to overcome this, improvements like adding fins, encasing PCMs, or utilizing high-conductivity materials like porous media have been investigated [161]. Utilizing nucleating chemicals or advanced encapsulation methods to address phase separation and supercooling problems. If there are problems with nanoparticle agglomeration in NePCMs, apply the right stabilizers or surfactants, optimize the synthesis and dispersion processes, and manage the storage conditions [162]. To stop corrosion and preserve system longevity, it is also essential to carefully choose PCM and containment materials to ensure material compatibility.

8. Comparative analysis and strategic research outlook

The application of PCMs in solar water heating systems has shown promising results across a range of experimental and numerical studies.

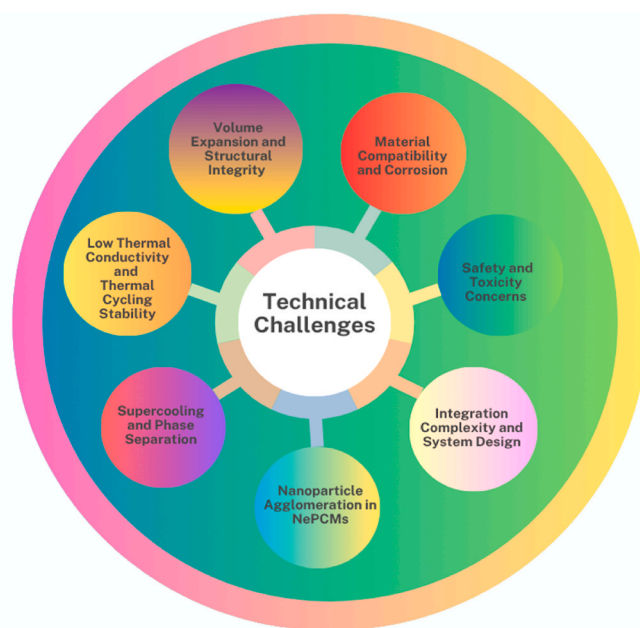


Fig. 13. Technical challenges of PCM/NePCM-integrated solar water heaters.

By comparing different PCM types, system configurations, and enhancement methods such as the use of nanoparticles, encapsulation techniques, and modified collector designs it is clear that PCMs can substantially improve thermal storage efficiency and system performance. Despite these advancements, challenges remain, particularly related to low thermal conductivity, long-term material stability, and integration at scale. To address these gaps, future research should focus on the development of hybrid or composite PCMs with improved thermal properties, along with innovations in system design to optimize heat transfer. Additionally, more real-world performance evaluations and standardized testing frameworks are needed to better assess technological readiness and guide the practical implementation of PCM-based solar thermal systems. The comparative analysis of experimental and numerical studies, as presented in Table 4, highlights the significant influence of PCM selection, system geometry, and material enhancements on the performance of solar desalination systems. Paraffin wax remains the most widely used PCM due to its favorable thermal properties, often demonstrating enhanced productivity when combined with design innovations or thermal conductivity enhancers like nanographene oxide and CuO nanoparticles. Such combinations have resulted in remarkable increases in freshwater yield up to 100 % improvement in some configurations. Additionally, altering system geometries, such as pyramidal stills, tubular stills with corrugated surfaces, and compound parabolic concentrators, further contributes to efficiency gains by optimizing solar absorption and thermal distribution. Organic PCMs like stearic, lauric, and palmitic acids also show competitive performance, with palmitic acid achieving up to a 30 % increase in daily output. Numerical simulations corroborate experimental findings, emphasizing the potential of dynamic PCM arrangements and molten salt mixtures in enhancing thermal performance. Overall, the data suggest that synergistic optimization of PCM type, thermal additives, and system architecture is key to maximizing freshwater production in solar stills.

Further, a comprehensive review of recent applications reveals the effective use of a wide variety of PCMs in diverse SWH configurations, including flat plate solar water heaters (FPSWH), evacuated tube collectors (ETCSWH), and parabolic trough systems. Organic PCMs such as paraffin wax, fatty acids, and beeswax dominate these applications due to their favorable melting points and chemical stability. In FPSWH systems, paraffin wax has achieved maximum daily efficiencies up to 65 %, while inorganic options like sodium acetate showed seasonal variations in performance, achieving 26–32 % efficiency. Enhancements through nano-additives such as CuO, SiC, and expanded graphite have

significantly improved thermal conductivity by up to 162.4 % in some cases and reduced charging/discharging time. Dual-PCM systems (e.g., tritriacontane and erythritol) have yielded up to 66 % efficiency gains compared to non-PCM setups. Notably, modifications such as incorporating nano-PCM in aluminum containers, or using PCMs in steel balls, have demonstrated prolonged thermal retention, keeping water temperatures significantly higher than ambient conditions for extended durations. Furthermore, exergy efficiency improvements of over 5 % have been reported in systems using nano-enhanced paraffin wax. Overall, the integration of PCM in SWH systems leads to enhanced energy storage capacity, improved efficiency, stabilized output temperatures, and reduced operational fluctuations, thereby reinforcing the value of PCM-based thermal energy storage for both domestic and centralized solar heating applications. These recent developments in SWH systems and their performance with PCMs have been compiled in Table 5.

Furthermore, an overview of the review conducted on the classification and application of a variety of PCMs in solar water heaters is illustrated in Table 6. The integration of various phase change materials (PCMs) into solar water heating (SWH) systems has been extensively explored to improve thermal storage efficiency and overall system performance. It summarizes the diverse range of PCMs spanning organic, inorganic, eutectic, and hybrid formulations applied across different types of SWH systems, particularly flat plate collectors (FPCs) and evacuated tube collectors (ETCs). These materials are strategically embedded at key locations such as the solar collector, storage tank, riser tubes, and insulated pipelines, allowing the latent heat storage property of PCMs to smooth out temperature fluctuations and extend thermal delivery beyond sunlight hours. To further enhance heat transfer performance, numerous thermal optimization techniques have been adopted. These include the incorporation of nanoparticles (e.g., Al_2O_3 , CuO, graphene) to increase thermal conductivity, the use of metal foams or extended surfaces like longitudinal, tree-shaped, or helical fins, and the encapsulation of PCMs at macro, micro, or nano scales to ensure structural stability and efficient melting-solidification cycles. Modifications in collector geometry, riser tube configuration, and storage tank design have also been shown to significantly impact heat transfer rates and PCM utilization. Collectively, these enhancements contribute to superior system metrics, such as higher latent heat storage capacity, improved charging/discharging durations, elevated thermal efficiency, and greater daily energy yield. Moreover, the improved energy conversion and storage efficiency lead to reduced auxiliary energy use and enhanced carbon mitigation, underlining the sustainability potential of

Table 4
Comparative studies on experimental parameters.

PCM Type	Nature of Work	System Geometry	Outcomes	Ref.
Paraffin wax with nano graphene oxide	Experimental	Tubular solar still (TSS)	TSS, TSS with PCM, and TSS with NPCM are reported to generate cumulative yields of 2.59, 3.35, and 5.62 kg per m ² area, respectively.	[163]
Paraffin Wax	Experimental	Pyramidal solar still	With PCM, productivity increases to 11.6 % and peak yield increases by 9.5 %.	[164]
Paraffin Wax, Stearic Acid, and Lauric Acid	Experimental	Passive Solar Still System with copper cylinder	The respective values for paraffin wax, stearic acid, and lauric acid were 2482, 2295, and 2210 ml/m ² -day.	[165]
Palmitic acid	Experimental	Single basin solar still with pin fins	Increased productivity to 5.4 L/m ² /day, which is a 30 % increase in daily output.	[166]
Paraffin wax	Experimental	Compound parabolic concentrator-concentric circular tubular solar still	Fresh water productivity increased by 8 % as a result of the 5330 ml/m ² /day fresh water output.	[167]
Paraffin wax	Numerical	Single slope solar still	A solar still with a moving layer of PCM produced 6.04 kg/m ² of freshwater daily.	[168]
Potassium Nitrate and Sodium Nitrate	Numerical	Parabolic trough collector	Increased the instantaneous power by 7 % and raised the outlet temperatures by 6–8 °C.	[169]
CuO nanoparticles and paraffin wax	Experimental	Wick solar still	The corrugated wick solar still's daily yields were 100 % higher than those of a traditional solar still.	[170]
Saline water	Experimental	Tubular solar still	Using a semicircular corrugated surface increased the desalination rate by 26.27 %.	[171]
Lauric acid	Experimental	Pyramidal solar still	The integration results in a 94.96 % increase in freshwater yield per day when compared to a traditional solar still.	[172]

Table 5

Recent developments of SWH with PCM for solar heat energy storage.

S. N.	SWH category	PCM type	TES material	Results	References
1	FPSWH	Organic	Paraffin wax	Maximum daily efficiency achieved is 65 %.	[173]
2	FPSWH	Inorganic	Sodium Acetate	Maximum efficiency in winter and summer found to be 26 % and 32 % respectively.	[174]
3	FPSWH	Organic	SiC and CuO nano composite with paraffin wax as PCM	Thermal conductivity of Nano composite PCM enhanced by 22.53 %.	[175]
4	ETSWH	Organic	Paraffin with nano CaO ₂ particle	Daily exergy and energy efficiency are 0.44 % and 10.89 % higher than without PCM respectively.	[176]
5	FPCSWH	Organic	Paraffin Wax with CuO nano particle (20 kg of paraffin wax is combined with 10 g of 40 nm nano CuO)	Highest outlet water temperature obtained is 80.6 °C.	[177]
6	FPCSWH/ETCSWH	Organic	Solid-Liquid Organic PCM	The suitable phase change temperature was discovered to be between 40 and 70 °C.	[178]
7	FPCSWH/ETCSWH	Organic	Paraffin	Only useful in low as well medium temperature applications.	[179]
8	FPCSWH	Organic	Paraffin Wax	Thermal energy storage density is improved by 39 %.	[180]
9	ETCSWH	Organic	Dual-PCM, Tritriacontane and Erythritol	66 % improvement in efficiency compared to without PCM.	[181]
10	Domestic Solar Water Heater	Organic	Low-temperature PCM like salt hydrates paraffins, fatty acids, and alcohols	Greater thermal efficiency and longer operational lifetime achieved using PCM in solar collector systems.	[182]
11	Solar Hot Water System Centralized	Organic	Paraffin wax with expanded graphite in Steel ball	In a centralized SWH, PCM with a 60 °C melting point is more efficient than 55 °C.	[183]
12	Solar Domestic hot water system	Organic	Three kilograms of paraffin, stearic acid (PS), palmitic acid (PP), and myristic acid (SM) mixed with an 80:20 wt percent ratio was created with 10 % (PCM weight) graphite	Paraffin with stearic acid (with 10 % graphite) provided the best results.	[184]
13	Solar water heating system (PCM in cylindrical aluminium container)	Organic	Paraffin wax	Water temperature in the storage tank remained at least 30 °C higher than the surrounding temperature for a 24-h test condition.	[185]
14	FPCSWH	Organic	Paraffin wax and paraffin wax with 20-nm nano-Cu particles (1.0 wt%) PCM nanocomposite	The maximum efficiency of 52 % has been obtained at 10° inclination of solar collector for PCM nanocomposite.	[137]
15	Glass evacuated tube solar water heater with NCPCM, PCM and without PCM	Organic	Paraffin wax with 1.0 mass% SiO ₂ nanoparticle	Exergy efficiencies are found to be 19.6 %, 22.0 %, and 24.6 % for without PCM, with PCM, and with NCPCM, respectively.	[186]
16	Parabolic trough collector types solar water heater with PCM	Organic non-paraffin PCM	Beeswax as PCM	Provided warm water at 40–60 °C for 5 h with Beeswax as PCM.	[187]
17	Heat piper-based ETCSWH	Organic	Paraffin wax	49.9 % improvement in efficiency.	[143]
18	SWH	Organic	Myristic acid	Myristic acid as PCM stabilized water temperature, reduced tank volume, and maintained 51–52 °C overnight.	[188]
19	SWH	Organic	Wax based PCM	60 % reduction in charging/discharging duration due to graphite addition.	[189]
20	SWH	Organic	Paraffin wax and nano enhanced paraffin wax	8.4 % improvement in efficiency.	[190]
21	SWH	Organic	Expanded graphite PCM	Thermal Conductivity of the composite PCM enhanced by 162.4 % compared to the pure alloy.	[191]
22	SWH	Organic	Expanded graphite mixed with stearic acid	63.3 % reduction in melting time of composite PCM.	[192]

Table 6

Classification and summary of the studies conducted.

Types of PCM	Application	Incorporation of PCM in SWH	Techniques to enhance thermal characteristics and performance of system with PCM	Effect of PCM on system performance as compared to conventional system
Organic, Inorganic, Eutectic, Mixture of PCM and Hybrid PCM	Solar water heating system (Flat plate collector based, Evacuated tube collector based)	PCM can be used in - Collector of heater - Storage tank - Around riser tube - Insulated pipe work.	- PCM in metal foam, addition of nanoparticle in PCM, addition of expended graphite, - Encapsulation (macro encapsulation microencapsulation, nano encapsulation) of PCM in storage tank. - Incorporation of fin where PCM's are used. - Incorporation of fin with PCM, - Modification in riser tube, evacuated tube collector, - Modification in storage tank	- Higher latent heat storage capacity, - Better charging/discharging pattern - Higher thermal efficiency - Higher heat transfer rate - Higher system daily efficiency - Higher CO₂ mitigation

PCM-integrated SWH systems. Thus, Table 6 reflects the critical role of material innovation and thermal design synergy in advancing solar water heating technologies for real-world applications.

9. Performance metrics and practical considerations

Several recent studies have highlighted the importance of system geometry and PCM type in improving the thermal response of solar water heating systems. Among various PCMs, paraffin wax has consistently shown better performance when paired with enhanced heat transfer structures. For example, introducing a combination of triangular and rectangular fins within the thermal storage cavity has led to a notable increase in charging speed, while round and longitudinal fins have similarly reduced both melting and solidification times. More complex fin designs such as dendritic or tree-shaped structures have proven even more effective. These configurations have significantly shortened the discharging durations in systems using lauric acid and water as PCMs, with some studies reporting over 50 % improvement in heat release time. Other geometric enhancements, like helical fins or specialized heat exchangers such as U-shaped and multi-tube configurations, have also shown substantial reductions in charging times. One of the most promising strategies involves combining metal foam with fins inside cylindrical shells, which has yielded dramatic reductions in both heat storage and release durations. Overall, the comparative findings suggest that carefully tailoring the internal geometry of PCM-based systems is key to enhancing thermal performance and ensuring more efficient, responsive solar water heating applications. A comparative studies of improvement in charging and discharging duration on solar

Table 7

Comparative studies of improvement in charging and discharging duration on solar water heating systems.

PCM Type	Geometry of system	Improvement in charging and discharging duration	Ref.
Paraffin wax	Hybrid triangular and rectangular fins in a cavity	The rate of charging increased by nearly 57 %.	[193]
Ternary $\text{Li}_2\text{CO}_3 + \text{K}_2\text{CO}_3 + \text{Na}_2\text{CO}_3$	Triangular-finned triplex tube	The maximum reduction in the system's discharging duration is 30.98 %.	[194]
Paraffin wax	Round-finned triple tube	34 % increase in the charging rate.	[195]
Paraffin wax	Shell with longitudinally finned tube	Compared to a finless design, the use of rectangular fins reduced the melting and solidification times by about 52 % and 44 %, respectively.	[196]
Water	Triplex tube with rectangular, tree-like fins	The solidification time was reduced by about 51.4 % when dendritic fins were used in place of rectangular fins.	[197]
Lauric acid	Tree-shaped fins on a shell and tube	The solidification and melting times were reduced by 66.2 % and 4.4 %, respectively, by dendritic fins.	[198]
Paraffin C-25	Helical fins on a cylindrical shell	At 35 fins, the charging and discharging processes can be decreased down by about 29.5 % and 20.9 %, respectively.	[199]
Paraffin RT-82	Multi-tube heat exchanger	The charging time has been reduced by 5234 s.	[200]
Paraffin RT-35	U-shaped heat exchanger	The charging time reduction of 91.79 %.	[201]
Paraffin	Fin-metal foam-filled cylindrical tube	With 61.6 % and 82 % reductions in heat storage and release times, respectively, the fin-foam metal structure exhibits the best heat storage/release performance.	[202]

water heating systems is explained in Table 7. These studies represent the improvement in charging and discharging duration on solar water heating system utilizing PCMs. The process of absorbing and releasing thermal energy during a phase transition is known as charging and discharging in a PCM. The PCM stores energy as latent heat when it melts and absorbs heat during charging. On the other hand, the PCM releases heat and solidifies during discharging, releasing the energy that has been stored. Increasing the heat transfer rate is essential for reducing the charging and discharging times of PCMs. In general, charging and discharging durations are influenced by PCM properties and system geometry.

The integration of phase change materials (PCMs) with innovative system enhancements has shown a clear positive impact on both the thermal efficiency and cost-effectiveness of solar water heating systems. Various studies highlight that combining PCMs with features such as rotating drums, external condensers, ultrasonic humidifiers, and photovoltaic-powered heaters can significantly improve daily efficiency. For instance, a solar still using saline water along with a rotating drum and copper oxide nanofluids achieved an impressive efficiency of 85.5 %, while a system incorporating paraffin wax with photovoltaic and Peltier components reached 80 %. In comparison, conventional systems typically operate at much lower efficiencies, ranging from 34 % to 50 %. In addition to performance gains, several modified systems also demonstrated cost advantages. A system utilizing paraffin wax with photovoltaic support reduced the cost of freshwater production to \$0.008/L, slightly lower than the \$0.009/L of the traditional setup. Similarly, a tray-type solar still improved not only in efficiency but also reduced production cost to \$0.021/L, compared to \$0.028/L in a conventional design. These findings collectively indicate that well-designed PCM-based solar water heating systems can deliver both improved thermal performance and economic viability, making them attractive solutions for sustainable energy applications. The summary of recent works on daily efficiency and cost analysis of solar water heating systems have been compiled in Table 8. The table represent overview of the daily efficiency and cost analysis of solar water heating systems. A solar heating system's daily efficiency is a measure of how well sunshine is converted into useful heat. The design of the system, the amount of sunlight present, and the surrounding temperature are some of the variables that affect this efficiency. Furthermore, the cost of the tested stills depends on the kind of still and its parts. This will help determine the energy savings and payback period of investing in a less energy-intensive system, which is probably going to cost more to purchase.

10. Conclusions

Phase Change Materials are emerging as a key enabler for improving the efficiency and sustainability of solar water heating systems. Their unique ability to store thermal energy as latent heat during phase transitions helps balance fluctuations in solar energy availability, enabling more consistent and reliable hot water supply. Among various PCM types, organic materials like paraffin wax are widely used for low-temperature applications due to their chemical stability and ease of integration. However, real-world implementation still faces challenges such as low thermal conductivity, flammability, material degradation, and high costs. Recent research and experimental insights offer encouraging solutions to these limitations. Incorporating nanoparticles, using extended surfaces such as fins, embedding PCMs within metal foams, and applying advanced encapsulation techniques have all shown to significantly boost heat transfer rates and shorten charging/discharging cycles. These enhancements not only improve system responsiveness but also increase overall energy conversion efficiency translating into reduced reliance on auxiliary energy sources and substantial carbon emission reductions.

Furthermore, thoughtful system design such as modified collector geometries, hybrid heat exchangers, and the strategic placement of PCMs has proven to enhance energy retention and delivery. The

Table 8

Summary of daily efficiency and cost analysis of solar water heating systems.

Type of PCM	Advancement in techniques	Improvement in Daily Efficiency	Cost Analysis	Ref.
Saline Water	Trays solar still with external bottom and Top mirrors (TSSIBTM)	The TSSIBTM achieved the highest efficiency of 50 % at a tray height of 0.5 cm, whereas the conventional solar still founded 34 %.	The conventional solar still and the tray solar still have fresh water production costs of 0.028 and 0.021 \$/L, respectively.	[203]
Al ₂ O ₃ based paraffin wax	Integrated external condenser and stepped distiller.	At 2000 rpm, the maximum daily efficiency was approximately 52.4 %.	Stepped still and reference still distilled water costs 0.014 and 0.015 \$/L, respectively.	[204]
Saline Water	An external condenser, a solar water heater, and CuO nanoparticles based Solar still with a rotating drum.	Using the external condenser, solar water heater, and nanofluids increased the drum still's efficiency to 85.5 % at a drum speed of 0.1 rpm.	Conventional and rotating drum solar stills with condenser, heater, and nanoparticles were estimated to cost 0.05 and 0.039 \$, respectively, for 1 L of distillate.	[205]
Paraffin wax	An evacuated solar collector and an ultrasonic humidifier combine in a single solar still.	The highest efficiency of 49 % and 42 % is achieved for water depths of 25 and 35 mm, respectively.	The cost of 1 L for freshwater productivity reaches 0.065 \$ and 0.037\$ for the conventional and modified solar still respectively.	[206]
Water	Dual-slope solar power plant with thermoelectric enhancement.	In comparison to a conventional solar still, the efficiency of a modified solar still has increased by 76.4 %.	Water costs for modified and standard solar stills were determined to be 0.1055 and 0.1763 dollars per lit/m ² , respectively.	[207]
Paraffin wax	Phase-change materials (paraffin wax), Peltier modules, and photovoltaic-powered electric heaters are integrated to a solar still.	At its maximum, the Modified Solar Still (MSS) attained a thermal efficiency of about 80 %, whereas the Conventional Solar Still (CSS) reached about 50 %.	The Modified Solar Still (MSS) cost \$0.008 per liter, while the Conventional Solar Still (CSS) cost \$0.009.	[208]
Stearic and Palmitic acids	Steel wool fiber (SWF) and binary eutectic PCM (a combination of stearic and palmitic acids) were integrated into the solar still.	The modified double slope solar still (PCM + SWF with a water depth of 2 cm) has an average energy efficiency of 41 %.	–	[209]
Stones and silica sand	PCM-equipped solar stills have hollow cylindrical pin fins.	The suggested system's thermal efficiency was increased by 37 %.	–	[210]

integration of PCMs with components like photovoltaic panels and ultrasonic humidifiers also indicates the growing potential for hybrid systems that blend thermal and electrical efficiencies. Such innovations open new avenues for scalable and climate-resilient solar technologies. Looking ahead, interdisciplinary collaboration across materials science, mechanical engineering, and environmental policy will be essential to unlock the full potential of PCMs in solar water heating. Advancements in cost-effective and eco-friendly PCMs, long-term durability, fire safety improvements, and modular system architectures will pave the way for widespread deployment. In essence, PCM-based solar water heating systems are not just an incremental improvement they represent a transformative step toward decentralized, clean energy solutions that can meet the growing demand for thermal comfort with minimal environmental cost. With continued innovation and practical integration strategies, PCMs could soon become a cornerstone in the global shift toward sustainable and resilient energy infrastructure.

11. Future scopes

The potential for further advances and broad implementation of solar water heaters utilizing PCMs is substantial. In order to advance this technology, it is necessary to thoroughly investigate and conduct research in numerous crucial areas:

- **Enhancing PCM Performance:** It is important to concentrate on enhancing the thermal conductivity, particularly in the case of organic versions. Advancements in nano composites, sophisticated encapsulation processes, and the integration of innovative materials have the potential to address current constraints, resulting in enhanced energy storage efficiency.
- **Development of Cost-Effective PCM Solutions:** The exorbitant expenses linked to specific PCMs continue to be a significant obstacle. Future research should focus on the advancement of economical alternatives that maintain high performance standards. One possible approach is to investigate novel materials, synthesis techniques, or large-scale production methods to enhance the economic feasibility of PCM-based solar water heaters.
- **Safety and Environmental Considerations:** It is imperative to prioritise addressing the safety concerns related to organic PCMs, particularly in terms of fire safety, in order to achieve wider acceptability. Future

research should focus on the development of ecologically sound and secure PCMs to guarantee the long-term viability of solar water heaters.

- **Integration of Advanced Materials:** Advanced materials, such as carbon nanotubes and nanoadditions, can significantly improve the performance of inorganic salt hydrates. Research in this field should investigate innovative methods to address difficulties like as super-cooling, phase segregation, and corrosiveness.

To summarise, the future potential of solar water heaters with PCMs requires a comprehensive approach that includes several fields such as material science, engineering, and environmental issues. Uninterrupted advancement and cooperative research endeavours can facilitate the development of PCM-based solar water heating systems that are more environmentally friendly, economically efficient, and generally embraced.

CRediT authorship contribution statement

Brihaspati Singh: Writing – original draft, Data curation, Conceptualization. **Ravi Shankar Rai:** Writing – review & editing, Supervision, Methodology. **Pankaj Yadav:** Visualization, Methodology, Formal analysis, Data curation. **Sambhrant Srivastava:** Methodology, Formal analysis, Data curation. **Chandmani Yadav:** Writing – review & editing, Supervision.

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Abbreviation Used

PCMs	PCMs
TES	Thermal Energy Storage
SWH	Solar Water Heater
FPCSWH	Flat Plate Collector Solar Water Heater
ETCSWH	Evacuated Tube Collector Solar Water Heater
NCPCM	Nano composite PCM
NCPCM	Nano Composite PCM
SWCNT	Single wall carbon nanotube
MWCNT	Multiwall carbon nanotube
HCNT	Hydrophilic carbon nanotube
EGPHE	Expanded graphite paraffin heat exchanger
LHTES	Latent Heat Thermal Energy Storage
ETSWH	Evacuated Tube Solar Water Heater

Data availability

Data will be made available on request.

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